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PETROLEUM TRAINING INSTITUTE, EFFURUN

ESM 113 — Past Questions & Materials

by Renance

Course: ESM 113

Institution: Petroleum Training Institute, Effurun

Papers in this pack: 2

Levels: ND1

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Paper 1 of 2 — ESM 113 PRACTICAL MANUAL

Level ND1 • First Semester • Handout

PETROLEUM TRAINING INSTITUTE
P.M.B 20, EFFURUN, DELTA STATE
NIGERIA.

PRACTICAL MANUAL

DEPARTMENT: ENVIRONMENTAL SCIENCE
MANAGEMENT TECHNOLOGY

PROGRAMME: NATIONAL DIPLOMA (ND 1)

COURSE TITLE: ENVIRONMENTAL SCIENCE
MANAGEMENT

COURSE CODE: ESM 113

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PRACTICAL 1

TITLE: Recognizing igneous rocks

AIM: To identify different types of
igneous rocks

OBJECTIVES: * Determine the appropriate mineral
composition of the igneous rocks
provided – characterized as felsic (Pink
or light gray coloured rocks), mafic
(Dark coloured rocks) and intermediate
(Mixture of dark and light coloured
rocks).

* Determine texture of the igneous
rocks given – crystal grain sizes.

MATERIALS: Granite, Basalt, Gabbro, Quartz,
Feldspar etc.

INTRODUCTION: Metamorphic rock can be explained as
a rock which formed as a result of
transformation of pre-existing rock due
to environmental factors such as
changes in temperature, pressure,
mechanical force and the instability of
chemical components involved.
Metamorphic rocks are classified on the
basis of texture (foliated and non-

foliated), grain-size and mineral composition.

ACTIVITY: * Hold each igneous rock provided to you and observe the crystal grain sizes.

* Identify mafic, felsic and intermediate with respect to igneous rock approximate size composition, shapes and grain-to-grain interrelationships.

* Sketch the igneous rock given to you and indicate the crystal grain sizes where appropriate.

EXERCISE:

1. Describe the following with respect to igneous rock formation; Phaneritic, Aphanitic, Porphyritic, Vesicular, Glassy, and Pyroclastic.
2. Explain the porosity and permeability of igneous rock.

PRACTICAL 2

TITLE: Recognizing sedimentary rocks

AIM: To identify different types of sedimentary rocks

OBJECTIVES: * Determine layering, grain size, grain sorting and angularity as applicable to Clastic form of sedimentary rock.

* Determine biologic detrital material and precipitated mineral accordingly.

MATERIALS: Sandstone, siltstone, shale, mudstone, conglomerate, Breccia, gypsum, coal, limestone and etc.

INTRODUCTION: Generally, sedimentary rocks are identified as Clastic, Biochemical and Chemical in accordance with their grain size, grain shapes, Grain composition, Biologic detrital material and the composition of precipitated materials found in it. If sedimentary rock is termed "Clastic" is being classified based on its grain size, grain shape and grain composition. Biochemically formed type of sedimentary rock is classified based on the composition of biologic detrital material present in it

while chemically formed is categorized on the basis of composition of precipitated minerals available.

Moreover, clastic sedimentary rocks are comprised of small portion of pre-existing rock, compressed and cemented to form new brand rock e.g. Sandstone, shale, Conglomerate, Breccia and etc.

Chemical sedimentary rocks are formed when mineral precipitated out of large body of water and eventually, the water is being evaporated and leave behind the minerals content.

Halite and dolostone are part of the example of chemically formed type of sedimentary rocks. Biochemical sedimentary rocks are formed when remnant of living thing such as shells fall down to the bottom of water and further, being compacted and solidified.

ACTIVITY: * Grasp each sedimentary rock provided to you and experienced the gritty texture where applicable.

* Carefully identify grain size, grain sorting, angularity and layering with respect to clastic form of sedimentary rock.

* Identify and sketch Clastic, Biochemical and Chemical form of Sedimentary rocks.

EXERCISE:

1. What are the importance of biologic detrital material, precipitated mineral, grain size, grain shape and grain composition towards identification of sedimentary rocks.
2. Explain porosity and permeability of sedimentary rock.

PRACTICAL 3

TITLE: Identification of metamorphic rocks.

AIM: How to identify metamorphic rocks

OBJECTIVES: * Identify the characteristic features of metamorphic rock e.g. sparking &

shine, bands of colors and shades, spots of reflective dots, arranged patterns and grain-like display.

* To determine texture, grain size and mineral composition of metamorphic rocks.

MATERIALS: Marble, slates, phyllite, Gneiss, schist, quartzite skarn, hornfels, Navaculite, Arsilute and etc.

INTRODUCTION: A rock is composed of one or more distinctive minerals. Therefore, considering the substance in which rock is made is an essential aspect of rock identification. This is achieved through appropriate recognition of major minerals found in given rock sample and this process is called mineralogy.

Another significant factor in terms of identification of rock is to appropriately interpret the texture appeared in a given rock sample. Texture is composed of grain size, shape and grain-to-grain interconnection between the minerals found in a particular rock.

ACTIVITY: * Carefully observe the rocks displayed on the work bench before you.

* Segregate foliated from unfoliated form of metamorphic rocks.

* Identify grain size and mineral composition.

* Identify and sketch the following characteristics of metamorphic rock;

- ☐ Sparking and shines
- ☐ Bands of colors and shade
- ☐ Spots of reflective dots
- ☐ Arranged patterns
- ☐ Grain-like display

* Observe and outline the difference between the following:-

- ☐ Quartzite
- ☐ Slate

☐ Marble

EXERCISE:

1. State how slate, quartzite and marble are formed.
2. Explain porosity and permeability of metamorphic rock.
3. What is the importance of sandstone, shale and limestone to metamorphic rock.

PRACTICAL 4

TITLE: Recognizing meteorological instruments

AIM: To be familiarized with the meteorological apparatus.

MATERIALS: Sialometer, Ceilometer, Luxmeter, Transmissometer, Pyranometer, Barometer, Hygrometer, wet & dry, evaporated, thermometer, Anemometer, Rain gauge, Wind vane and etc.

ACTIVITY: * Kindly recognize the meteorological instruments displayed on the work bench before you.

* Draw and state the function of each instrument given.

* Calculate the total average rainfall per day (in inches) of the 7 samples given to you and comment on the result obtained.

RAINFALL RECORDS: Day 1 = 1.5, Day 2 = 1.8, Day 3 = 2, Day 4 = 1.7, Day 5 = 1.5, Day 6 = 1.5, Day 7 = 1.8.

EXERCISE:

Describe step by step method on how to use the following meteorological instruments:-

- (i) Anemometer
- (ii) Rain gauge
- (iii) Thermometer

PRACTICAL 5

The operation of evaporimeter

AIM / OBJECTIVE:

How to operate evaporimeter

PROCEDURES:

- ☐ Mount the evaporimeter on stand 1meter above sea level.
- ☐ Fill the pan with fresh water to reach the designated mark.

□ Carryout calibration: This is done by checking the functionality ultrasonic and temperature sensors.

DAILY OPERATION

□ Check water level once a day by measuring the level of water using dipstick or ruler.

□ Plug the evaporimeter to electric power supply.

□ Take the water level and temperature readings.

□ Take note of the time when readings are taken.

□ Unplug the equipment after taking readings.

□ Routine readings should be taken at the interval of 24hrs which is conducted each time by plugging the evaporimeter to electric power supply.

Calculate the pan evaporation using the following equation;

$$E_{pan} = A \times (D_0 - D) \times \rho$$

Where E_{pan} is the volume of water evaporated in M^3 , D_0 is the initial distance in m, D is the final distance in m, ρ is water density in kg/m^3 and A is the water surface area in m^2

Use the E_{pan} value to calculate

E_{t0} using the following equation

$$E_{t0} = E_{pan} \times 0.67$$

EXERCISE

Calculate the pan evaporation and E_{t0} of the following data:

$D = 2m$, $A = 14m^2$, $D_0 = 5m$, $\rho = 997kg/m^3$

Paper 2 of 2 — ESM 113 INTRODUCTION TO GEOGRAPHY

Level ND1 • First Semester • Handout

ESM 113 INTRODUCTION TO GEOGRAPY

GOAL: This course is designed to exposed students to basic geographic principles and their influence on physical earth forms.

GENERAL OBJECTIVE:

On completion of this course, the student should be able to;

0.1- Understand the structure that are involved in surface and subsurface environmental studies.

0.2- Understand processes that form Igneous rocks.

0.3- Understand processes that form Metamorphic rocks.

0.4- Understand processes that form Sedimentary rocks.

0.5- Understand the porosity and permeability of the rocks.

0.6- Understand the types of aquifers as geological formations Based on their physical properties.

0.7- Understand and interpret a table of formation.

0.8- Understand the difference between confirmed and non-confirmed aquifers.

0.9- Understand the effects of non-conformities on ground water movement.

1.0 Understand the processes that form geological structures and their effects on surface water and ground water.

1.1 Understand how to subsurface relationship developed between rock units and their influence on ground water flow.

1.2 Understand basic climatological factors.

0.1-Understand the science that are involved in surface and subsurface environmental studies.

UNIT STRUCTURE:

1.1- Define geography.

1.2- Explain the relationship of geography with the environmental

1.3- Explain the branches of the geography based on the subject matter and application.

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1.4- List types of data that are required in the field of environmental science and management.

1.1-Define geography.

INTRODUCTION:

Geography is the primitive earth science and its roots date back in the works of the early Greek scholars. The word “geography” was used by the Greek scholars Eratosthenes in (276-194BC). He was the first scholar to determine the circumference of the earth. It was coined from the two Greek words. Geo “Earth” and Graphein” to describe” literally meaning of geography is to describe about the earth’s surfaces. So, geography can simply be defined as the description of the earth. Hence, it better defined as the study of people, their activities, places, and physical things within the earth. In other word geography deals with the study of different people in different locations of the earth surface including their activities. Like agriculture. Mining, fishing, trading, manufacturing, and construction. It is also involving the study of places like USA, CHINA, NIGERIA, KANO, SOUTH AFRICA and GHANA and physical things within the earth such as rocks, oceans, mountains, plains, rivers, climate, vegetations, soil atmosphere etc. Furthermore, Geography is largely the study of the interaction of all physical and human phenomena and landscapes created by such interactions. It is about how, why, and where human and natural activities occur and how these activities are interconnected. Geography has undergone changes in its approach the earlier geographers were descriptive geographers, later geography came to be developed as an analytical science. Today the discipline is not only concerned with descriptive but also with analysis as well as prediction.

Geographical work in ancient Greece had followed two (2) distinct traditions. One was the mathematical tradition which was focused on fixing the location of places on the earth’s surface, and the other was gathering geographic information through travels and field work, according to them, the purpose of geography was to provide a description of the physical features and conditions in different parts of the world. The emergence of regional approach in geography also emphasized the descriptive character of geography. According to Humboldt,

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geography is the science related to nature and its studies and describes all material things found-on earth. Another important school of thought defined geography as the study of man-environment relationship.

IN THE ESSENCE, GEOGRAPHY IS CONCERNED WITH THE;

- Location
- Spatial and distribution of living and non-living things patterns and relations
- Regional characteristics of these elements on human life and settlement
- The forces that change the earth's features and processes
- Preparation of thematic of maps for planning and development
- Analyses of satellite images for various purposes
- Preparation and updating of maps for different applications.

DEFINITIONS OF GEOGRAPHY BY DIFFERENT

SCHOLARS:

1. According to Vidal De La Blanche: Geography is the science of places, concerned with the qualities of potentialities of countries. The character of a country is expressed by the totality of its features, the social diversities associated with the diversities of places.
2. According to American College Dictionary: Geography is the study of the areal differentiation of the earth surface, as shown in character, arrangement, and inter-relations over the earths elements such as climate, relief, soil, vegetation, population, land use, industries, or states and of the unit areas formed by the complex of these individual elements.
3. According to Richard Hartshorne, 1959: Geography provides accurate, orderly, and rational description and interpretation of the variable character of the earth surface.

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It is that discipline that seeks to describe and interpret the variable character from place of the earth as the world of man.

4. According to Strabo, 1970: Geography is closely related to its location, specific characteristics, and interrelationships of the region as a whole. This opinion was later called the concept of natural attribute of place.
5. According to Ptolemy, 150 CE: the purpose of geography is to provide a view of the whole earth by mapping the location of places.
6. According to Immanuel Kant, 1780: defined geography as a synoptic discipline synthesizing findings of other sciences through the concept of area or space.
7. According to Alexander Von Humboldt, 1845: geography

is a synthesizing discipline that connects the general with the special through measurement, mapping, and a regional emphasis.

SCOPE OF GEOGRAPHY

Geography has now acquired the status of science that explains the arrangements of various natural and cultural features on the earth surface. Geography is a holistic and interdisciplinary field of study engaged in understanding the changing spatial structure from past or future. Thus, the scope of geography is in various disciplines, like armed services, environment management, water resources, disaster management, meteorology and planning and various social sciences. Apart from that, a geography can help in day-to-day life like tourism, commuting, housing, and health related activities.

IMPORTANCE OF GEOGRAPHY

Geography is very important because of the following reasons;

1. To study the way of life of other people: it enables us to study the way of life of other people in other parts of the world. Without geography, it would have been difficult to know the existence and way of life of the Americans, Britain, the Brazilians and Japanese.

2. To study the physical environment around us: it enables us to study the physical features around us such as soil, vegetation, climate, rivers, oceans, and mountains.

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3. To know the importance of the physical features around us: it enables us to know the importance of the physical features around us and how they can be harnessed for human use. E.g., the climate of an area enables us to know the type of crops to grow; vegetation provides timber for us; rivers help in transportation, generation of electricity and provide fish for food.

4. To understand other areas of other subjects: It enables us to understand other areas of other subjects like vegetation and soil in agriculture, ecology and food chain in biology, population, industries, and trade in economics.

5. To choose a professional career: it enables us to choose a career from which we can earn a living and contribute to the economic development of our country. Careers like land surveying, environmentalist, urban and regional planning, estate management, military studies, aviation, architecture

will not be possible to study in higher institutions of learning without the knowledge of geography.

6. To study the social environment around us: geography helps us to study the social environments are schools, churches, mosques, market, cinema studio and etc.

7. It aids in internal and external trade: geography has helped us to know the resources of other regions or countries, and this has assisted us to determine both internal and external trade.

8. International relationship: geography in relation to other subjects like government, commerce and economics has assisted many countries, to have bilateral international relations with other countries of the world.

1.2- Explain the relationship of geography with the environmental

MAN-ENVIRONMENT RELATIONSHIPS AND THEIR IMPACTS ON EACH OTHER,

There is a considerable noticed that the earth's surface is ever changing; in general, the natural phenomena like mountains, rivers, lakes etc. change slowly while the cultural elements like buildings, roads, crops. Change fast. Travelling from one place to another you notice that the trees number and types of trees change from area to area. All this is because of the continuous interaction between the

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environment in which we live in and the way we use it. The study of geography is about observing such patterns. Another aspect of geography is to understand the factors or reason behind areal differentiation, how do social, cultural, economic, and demographic factors change our physical landscape and create new or altered landscapes by human activities. For example, human settlements area transformation of forest or barren land for living purpose by human being.

Geography studies the locations of the activities. Carefully identify patterns using maps and find out the reasons for these patterns. The areas are then described based on the distribution of landforms, population, house type and Agriculture. Such information is of benefit to the society. Such mapmakers are in great demand today. Nowadays, geographers, engineers, environmental scientists, city

planners, social scientists and many others learn to use GIS to understand the earth better.

Today all over the world there are problems related to proving food security, health, effective energy use and environmental conservation. Equally important are equality issues and sustainable development, all these can be achieved by using our resources in sustainable ways. Study of geography is, therefore, necessary to learn more about environmental processes and to understand how land use planning can help us to overcome such environmental related problems. The environment of an organism includes physical properties, which can be described as the sum of local abiotic factors such as insolation (sunlight), climate and geology, and biotic ecosystem, which includes other organisms that share its habitat.

Environment is defined as the surroundings or conditions where a man, plant and animal live or strives. In other words, the environment includes all the factors, external and internal. Living and non-living factors which affect an organism (both plants and animals). Surrounding of an organism, including both abiotic factors (e.g., Temp, rainfall) and biotic factors (e.g., predation, competition).

TYPES OF ENVIRONMENTS

□ **PHYSICAL ENVIRONMENT:** the biophysical environment is the biotic and abiotic surroundings of an organism or population, and they include the factors that have an influence on their survival, development and evolution. The

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physical environment includes plants, animals, climatic and factors such as temperature, wind, sunlight, soil, and rocks.

□ **SOCIAL ENVIRONMENT:** the social environment is the immediate physical and social surroundings that people live or in which something happens or manifests. The social environment also called non-physical environment relates to the environment created by human activities in changing the phase of landforms. It includes built environment in terms of schools, houses, roads, airports, seaports, playgrounds, stadia, markets, banks, shops and farms.

□ **CULTURAL ENVIRONMENT:** the cultural environment refers to the cultural and social aspect of a given society from its very nature. Their norms and beliefs that make up

the cultural environment. The cultural environment includes the religious activities in churches, mosques, way of dressing, our hair styles and ways of feeding.

□ HUMAN DIMENSIONS OF GLOBAL ENVIRONMENTAL CHANGES

- Greenhouse gas emissions — Coal-fired power station • Carbon dioxide • Methane • Fluorinated gases
- Human population — Biocapacity • climate change • Carrying capacity • Exploitation • Industrialization • I = PAT • Land degradation • Land reclamation • Optimum population • Overshoot (population) • Population density • Population density • Population dynamics • Population growth • Projections of population growth • Total fertility rate • Urbanization • Waste • Water conflict • Water scarcity • Over drafting
- Hydrology — Environmental impacts of reservoirs • Tile drainage • Hydrology (agriculture) • Flooding • Landslide
- Intensive farming — Agricultural subsidy • Barn fires • Environmental effects of meat production • Intensive animal farming • Intensive crop farming • Irrigation • Monoculture • Nutrient pollution • Overgrazing • Pesticide drift • Plasticulture • Slash-and-burn • Tile drainage • Zoonosis
- Land use — Built environment • Desertification • Habitat fragmentation • Habitat destruction • Land degradation • Land pollution • Lawn-environmental concerns • Trail ethics • Urban heat island • Urban sprawl
- Nanotechnology — Impact of nanotechnology
- Natural disasters
- Nuclear issues — Nuclear fallout • Nuclear meltdown • Nuclear power • Nuclear weapons • Nuclear and radiation

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accidents • Nuclear safety • High-level radioactive waste management

- Ocean trash — Garbage patch • Ghost net • Washed Ashore
- Water pollution

□ EFFECTS

- Climate change — Global warming • Global dimming • Fossil fuels • Sea level rise • Greenhouse gas • Ocean acidification • Shutdown of thermohaline circulation •

Environmental impact of the coal industry • Urban heat islands • Flooding

□ Environmental degradation — Loss of biodiversity • Habitat destruction • Invasive species

□ Environmental health — Air quality • Asthma • Birth defect • Developmental disability • Endocrine disruptors • Environmental impact of the coal industry • Environmental impact of nanotechnology • Electromagnetic field • Electromagnetic radiation and health • Indoor air quality • Lead poisoning • Leukemia • Nanotoxicology • Nature deficit disorder • One Health • Sick Building Syndrome • Environmental impact of hydraulic fracturing

□ Environmental issues with energy — Environmental impact of the coal industry • Environmental impact of the energy industry • Environmental impact of hydraulic fracturing

□ Environmental impact of transport

□ Environmental impact of aviation

□ Environmental impact of the petroleum industry — Exhaust gas • Waste tires • Motor vehicle emissions and pregnancy • Externalities of automobiles

□ Environmental impact of shipping (Cruise ships in Europe • Cruise ships in the United States)

□ Environmental issues with war — Agent Orange • Depleted uranium • Military Superfund site (Category only) • Scorched earth • War and environmental law • Unexploded ordnance

□ Overpopulation — Burial • Overpopulation in companion animals • Tragedy of the commons • Gender Imbalance in Developing Countries • Sub-replacement fertility levels in developed countries

□ Mutation breeding — Genetic pollution

□ Synthetic biology — Synthetic DNA • Artificially Expanded Genetic Information System • Hachioji DNA

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□ Genetically modified food — Genetically modified crops • Genetically modified livestock • Genetically modified food controversies

□ Pollution — Nonpoint source pollution • Point source pollution

□ Air pollution — Environmental impact of the coal industry

- Environmental impact of hydraulic fracturing • Indoor air quality • Smog • Tropospheric ozone • Volatile organic compound • Atmospheric particulate matter CFC • Biological effects of UV exposure Light pollution • Visual pollution Noise pollution
- Soil pollution — Alkali soil • Brownfield • Residual sodium carbonate index • Soil conservation • Soil erosion • Soil contamination • Soil salination • Superfund • Superfund sites
- Water pollution — Acid rain • Agricultural runoff • Algal bloom • Environmental impact of the coal industry • Environmental impact of hydraulic fracturing • Eutrophication • Fish kill • Groundwater pollution • Groundwater recharge • Marine debris • Marine pollution • Mercury in fish • Microplastics • Nutrient pollution • Ocean acidification • Ocean dumping • Oil spills • Soda lake • Ship pollution • Thermal pollution • Urban runoff • Wastewater Space debris • Interplanetary contamination • Ozone depletion
- Resource depletion — Exploitation of natural resources • Over drafting (groundwater) • Overexploitation
- Consumerism — Consumer capitalism • Planned obsolescence • Over-consumption
- Fishing — Blast fishing • Bottom trawling • Cyanide fishing • Ghost nets • Illegal, unreported and unregulated fishing • Overfishing • Shark culling • Shark finning • Whaling
- Logging — Clearcutting • Deforestation • Illegal logging
- Mining — Acid mine drainage • Environmental impact of hydraulic fracturing • Mountaintop removal mining • Slurry impoundments
- Water (depletion) — Anoxic waters • Aral Sea • California Water Wars • Dead Sea • Lake Chad • Water scarcity
- Toxicants — Agent Orange • Asbestos • Beryllium • Bioaccumulation • Biomagnification • Chlorofluorocarbons
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- (CFCs) • Cyanide • DDT • Endocrine disruptors • Explosives • Environmental impact of the coal industry • Herbicides • Hydrocarbons • Perchlorate • Pesticides • Persistent organic pollutant • PBBs • PBDEs • Toxic heavy

metals • PCB • Dioxin • Polycyclic aromatic hydrocarbons
 • Radioactive contamination • Volatile organic compounds
 ☐ Waste — Electronic waste • Great Pacific Garbage Patch •
 Illegal dumping • Incineration • Litter • Waste disposal
 incidents • Marine debris • Medical waste • Landfill •
 Leachate • Toxic waste • Environmental impact of the coal
 industry • Exporting of hazardous waste

☐ MITIGATION/SOLUTIONS

☐ Mitigation of aviation's environmental impact –
 Aviation taxation and subsidies (Air passenger tax •
 Kerosene tax) • Electric aircraft • High-speed rail
 ☐ Conservation
 ☐ Ecosystems — Anoxic waters • Biodiversity • Biosecurity
 • Coral bleaching • Black carbon • Edge effect • Habitat
 destruction • Organic farming • Habitat fragmentation •
 In-situ leach
 ☐ Fishing — Blast fishing • Bottom trawling • By-catch •
 Cetacean bycatch • Gillnetting • Illegal, unreported and
 unregulated fishing • Environmental effects of fishing •
 Marine pollution • Overfishing • Whaling
 ☐ Forests — Clearcutting • Deforestation • reforestation
 and afforestation • Illegal logging • Trail ethics
 ☐ Natural resources — Resource depletion • Exploitation
 of natural resources • Steady-state economy • Waste
 hierarchy
 ☐ Species — Endangered species • Genetic diversity •
 Habitat destruction • Holocene extinction • Invasive
 species • Poaching • Pollinator decline • Species
 extinction • Threshold host density • Wildlife trade •
 Wildlife disease
 ☐ Energy conservation — Alternatives to car use •
 Efficient energy use • Car free city • Energy hierarchy •
 Local food
 ☐ Renewable energy — Renewable energy
 commercialization
 ☐ Recreation — Protected areas

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☐ Water conservation
 ☐ Disaster mitigation
 ☐ Environmental law — Environmental crime •
 Environmental justice • Polluter pays principle •

Precautionary principle • Regulatory capture • Trail ethics

□ Phase-out of fossil fuel vehicles

I. Environmental aspects of the electric car

II. Hydrogen economy

III. Rail electrification

IV. Scrappage program

V. Vehicle recycling

VI. Phase-out of single-use plastics

□ Phase-out of lightweight plastic bags (Australia • United States) • Biodegradable bags • Reusable shopping bag • Shopping trolley (caddy)

I. Bottled water ban — Reuse of bottles

II. Plastic straw ban

III. Sustainable agriculture

IV. Nutrition transition — Cellular agriculture (cultured meat) • Plant-based diet (reductionism • veganism • vegetarianism)

3 Explain the branches of the geography based on the subject matter and application

Research and careers in geography today are highly multidisciplinary. Geography is often organized into three branches: human geography, physical geography, and technical geography.[14][1] Geographers rarely focus on just one of these topics, often using one as their primary focus and then incorporating data and methods from the other branches. Geographers are more likely to identify closely with a specific branch when describing themselves to lay people than with grand traditions.

□ Physical geography is concerned with the study of processes and patterns in the natural environment like the atmosphere, hydrosphere, biosphere, and geosphere.

Physical geography (or physiography) focuses on geography as an Earth science.[36][37][38] It aims to understand the physical problems and the issues of lithosphere, hydrosphere, atmosphere, pedosphere, and global flora and fauna patterns (biosphere). Physical geography is the study of earth's seasons, climate,

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atmosphere, soil, streams, landforms, and oceans.[39]

Physical geographers will often work in identifying and monitoring the use of natural resources.

Physical geography can be divided into many broad categories, including:

- Biogeography
- Geomorphology
- Oceanography
- Climatology & metrology
- Environmental management
- Glaciology
- Hydrology & hydrography
- Paleogeography
- Pedology
- Landscape ecology.
- Coastal geography
- Human geography is concerned with studying people and their communities, cultures, economies, and interactions with the environment by studying their relations with and across space and place.

Human geography (or anthropogeography) is a branch of geography that focuses on studying patterns and processes that shape human society.[40] It encompasses the human, political, cultural, social, and economic aspects. In industry, human geographers often work in city planning, public health, or business analysis.

Human geography can be divided into many broad categories, such as:

- Cultural geography
- Political geography
- Social geography
- Economics geography
- Population geography
- Regional geography
- Religion geography
- Development geography
- Health geography
- Historical geography
- Transportation geography
- Urban geography
- Tourism geography

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□ Technical geography is interested in studying and applying techniques and methods to store, process, analyze,

visualize, and use spatial data. These branches use similar geographic philosophies, concepts, and tools and often overlap significantly.

Technical geography concerns studying and developing tools, techniques, and statistical methods employed to collect, analyze, use, and understand spatial data.[14][41]

Technical geography is the most recently recognized of the branches. As academic fields increasingly specialize in their nature, technical geography has emerged as a branch of geography specializing in geographic methods and thought.

[14] The emergence of technical geography has brought new relevance to the broad discipline of geography by serving as a set of unique methods for managing the interdisciplinary nature of the discipline. While human and physical geographers use the techniques employed by technical geographers, technical geography is more concerned with the fundamental spatial concepts and technologies than the nature of the data.[14] A technical geographer might work as a GIS analyst or create general reference maps incorporating human and natural features.

Technical geography can be divided into many broad categories, such as:

- Cartography
- Geoinformatics
- Geographical information science
- Remote sensing
- Geomatic
- Geostatistics
- Geographical information system
- Geovisualization
- Global positioning system
- Photogrammetry
- Spatial analysis

1.3- List types of data that are required in the field of environmental science and management

□ SPHERE OF THE ENVIRONMENT;

There are four major spheres of the environment.

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These are; Lithosphere, Hydrosphere, Atmosphere and Biosphere.

□ LITHOSPHERE: this is the solid portion of the earth,

which represent 30% of the earth surface, it forms the landmass of about 10-50km thick, it also called the outermost layer or ozone of the earth crust which made up of rocks and mineral materials like gravel, sand and soils.

□ IMPORTANCE OF LITHOSPHERE TO MAN;

□ Lithosphere forms the basis of all human settlement

□ It aids transportation through construction of roads, railways and airport,

□ All farming activities are carried out on lithosphere

□ All mineral resources like mining and explorations are derived from the same sources

□ HYDROSPHERE: this is the liquid portion of the earth, it covers about 70% of the earth's crust, including the ocean water (salty) account for about 97% of the hydrosphere. It holds water in forms of solid e.g., ice, liquid like water and gases like water vapour, and also includes all forms of fresh water like lakes, rivers and water vapour.

□ IMPORTANCE OF HYDROSPHERE TO MAN:

□ Hydrosphere provides water for domestic uses, e.g., washing, drinking and cooking.

□ It provides a medium for sport and transportation e.g., swimming and diving, seaport and shipping and etc.

□ It provides water for agricultural and industrial uses

□ It provides sea food and employment opportunity to fishermen.

□ It can also be used to generate electricity e.g., hydroelectric power.

□ It also serves as tourist Centre e.g., rivers and beaches.

□ ATMOSPHERE: this is the gaseous portion of the earth crust; it is the layer of gases surrounding the earth with over 99% lies within 30km of the earth surface. It contains 78% Nitrogen, 21% Oxygen, 0.03% Carbon dioxide and 0.97% rare or inert gases. The weather phenomenon such as cloud, rain and snow are largely confined in the

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atmosphere and its density decreases progressively with elevation.

□ IMPORTANCE OF ATMOSPHERE TO MAN:

□ The atmosphere is the habitat of some living organism.

□ It provides oxygen for plants and animals for respiration.

- It provides carbon dioxide for photosynthesis by plants.
- It provides oxygen for combustion or burning.
- It acts as green house and provides nitrogen for plants (protein synthesis).
- It contains ozone layer which acts as protective cover against the burning effects of ultra violet rays of the sun.
- BIOSPHERE: this is the zone of the earth occupied by living organisms; it is a layer of life which exists on earth surface and the lithosphere with a maximum thickness of only few kilometers. It is a narrow zone where complex biological and chemical activities occur including organisms like plants, animals and microorganisms.
- IMPORTANCE OF BIOSPHERE TO MAN:
 - Plants in the biosphere provides food for man.
 - They also provide a source of energy like firewood for man.
 - It provides shelter in form timber for man.
 - It provides raw materials like timber hides and skin, cotton for industries.
 - Activities within the biosphere provide employment for man.
 - It aids the balancing and purification of atmospheric gases e.g., carbon dioxide during respiration and oxygen during photosynthesis.

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0.2 -Understand processes that form Igneous rocks

2.1 Define rock

A rock is a natural substance composed of solid crystals of different minerals that have been fused together into a solid lump. These minerals may or may not have been formed at the same time. What matters is that natural processes glued them all together.

Furthermore, a rock is a mass of natural deposit forming the solid part of the earth's crust. It may be hard like granite or soft like a clay or sand. A rock is in fact, not a chemical compound but an aggregate of a large number of minerals. In some instances, one particular mineral may be dominant in a rock. Such a rock is called by the name of that particular mineral and it is considered its ores. Bauxite and hematite ore, examples of such rocks that are ores of aluminum and iron, respectively. Geology is the study of Earth and its components, including the study of rock formations.

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□ Petrology is the study of the character and origin of rocks.

□ Mineralogy is the study of the mineral components that create rocks.

The study of rocks and their components has contributed to the geological understanding of Earth's history,

□ the archaeological understanding of human history, and the development of engineering and technology in human society.[1]

While the history of geology includes many theories of rocks and their origins that have persisted throughout human history, the study of rocks was developed as a formal science during the 19th century. Plutonism was developed as a theory during this time, and the discovery of radioactive decay in 1896 allowed for the radiocarbon dating of rocks. Understanding of plate tectonics developed in the 20th century.[2]

The Grand Canyon is an incision through layers of sedimentary

rocks.

In geology, rock (or stone) is any naturally occurring solid mass or aggregate of minerals or mineraloid matter. It is categorized by the minerals included, its chemical composition, and the way in which it is formed. Rocks form the Earth's outer solid layer, the crust, and most of its interior, except for the liquid outer core and pockets of magma in the asthenosphere. The study of rocks involves multiple subdisciplines of geology, including petrology and mineralogy. It may be limited to rocks found on Earth, or it may include planetary geology that studies the rocks of other celestial objects.

1. 2.2 Describe types of rock

Rocks are usually grouped into three main groups: igneous rocks, sedimentary rocks and metamorphic rocks. Igneous
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rocks are formed when magma cools in the Earth's crust, or lava cools on the ground surface or the seabed. Sedimentary rocks are formed by diagenesis and lithification of sediments, which in turn are formed by the weathering, transport, and deposition of existing rocks. Metamorphic rocks are formed when existing rocks are subjected to such high pressures and temperatures that they are transformed without significant melting.

Humanity has made use of rocks since the earliest humans. This early period, called the Stone Age, saw the development of many stone tools. Stone was then used as a major component in the construction of buildings and early infrastructure. Mining developed to extract rocks from the Earth and obtain the minerals within them, including metals. Modern technology has allowed the development of new man-made rocks and rock-like substances, such as concrete. Rocks are composed primarily of grains of minerals, which are crystalline solids formed from atoms chemically bonded into an orderly structure.[4]: 3 Some rocks also contain mineraloids, which are rigid, mineral-like substances, such as volcanic glass,[5]: 55, 79 that lacks crystalline structure. The types and abundance of minerals in a rock are determined by the manner in which it was formed. Most rocks contain silicate minerals, compounds that include silica tetrahedra in their crystal lattice, and account for about one-third of all known mineral species and about 95% of the earth's crust.[6] The proportion of silica in rocks and minerals is a major

factor in determining their names and properties. [7]

Rock outcrop along a mountain creek near Orosí, Costa Rica.

Rocks are classified according to characteristics such as mineral and chemical composition, permeability, texture of the constituent particles, and particle size. These physical properties are the result of the processes that formed the

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rocks.[5] Over the course of time, rocks can be transformed from one type into another, as described by a geological model called the rock cycle. This transformation produces three general classes of rock: igneous, sedimentary and metamorphic. Those three classes are subdivided into many groups. There are, however, no hard-and-fast boundaries between allied rocks. By increase or decrease in the proportions of their minerals, they pass through gradations from one to the other; the distinctive structures of one kind of rock may thus be traced, gradually merging into those of another. Hence the definitions adopted in rock names simply correspond to selected points in a continuously

IGNEOUS ROCK

Sample of igneous gabbro

Igneous rock (derived from the Latin word igneus, meaning of fire, from ignis meaning fire)[9] is formed through the cooling and solidification of magma or lava. This magma may be derived from partial melts of pre-existing rocks in either a planet's mantle or crust. Typically, the melting of rocks is caused by one or more of three processes: an increase in temperature, a decrease in pressure, or a change in composition.[10]: 591–599

CHARACTERISTICS OF IGNEOUS ROCK

1. Igneous rocks occur in a wide range of geological settings: shields, platforms, orogens, basins, large igneous provinces, extended crust and oceanic crust
2. Igneous rocks are crystalline in structure, i.e., they contain crystals.
3. They do not occur in layers i.e.; they are non-stratified rocks.
4. They do not contain fossils.
5. They are usually hard and impervious.
6. They are either acidic or basic.

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7. They vary in color (light or dark)

8. Crystal's size varies from specks to glassy mass

□ FORMATION OF IGNEOUS ROCK

Igneous rocks are formed through the cooling and solidification of magma or lava. The magma can be derived from partial melts of existing rocks in either a planet's mantle or crust. Typically, the melting is caused by one or more of three processes: an increase in temperature, a decrease in pressure, or a change in composition. Solidification into rock occurs either below the surface as intrusive rocks or on the surface as extrusive rocks. Igneous rock may form with crystallization to form granular, crystalline rocks, or without crystallization to form natural glasses. Igneous rocks occur in a wide range of geological settings: shields, platforms, orogens, basins, large igneous provinces, extended crust and oceanic crust.

Formation of igneous rock

Volcanic eruptions of lava are major sources of igneous rocks. (Mayon volcano in the Philippines, erupting in 2009)

GEOLOGICAL SIGNIFICANCE: Igneous and metamorphic rocks make up 90–95% of the top 16 kilometers (9.9 mi) of the Earth's crust by volume. [1] Igneous rocks form about 15% of the Earth's current land surface. [note 1] Most of the Earth's oceanic crust is made of igneous rock.

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Igneous rocks are also geologically important because:

□ their minerals and global chemistry give information about the composition of the lower crust or upper mantle from which their parent magma was extracted, and the temperature and pressure conditions that allowed this extraction;[3]

□ their absolute ages can be obtained from various forms of radiometric dating and can be compared to adjacent geological strata, thus permitting calibration of the geological time scale;[4]

□ their features are usually characteristic of a specific tectonic environment, allowing tectonic reconstructions (see plate tectonics);

□ in some special circumstances they host important mineral

deposits (ores): for example, tungsten, tin,[5] and uranium[6] are commonly associated with granites and diorites, whereas ores of chromium and platinum are commonly associated with gabbros.[7]

IGNEOUS ROCKS ARE DIVIDED INTO TWO MAIN CATEGORIES:

- Plutonic or intrusive rocks result when magma cools and crystallizes slowly within the Earth's crust. A common example of this type is granite.
- Volcanic or extrusive rocks result from magma reaching the surface either as lava or fragmental ejecta, forming minerals such as pumice or basalt.[5] Magmas tend to become richer in silica as they rise towards the Earth's surface, a process called magma differentiation. This occurs both because minerals low in silica crystallize out of the magma as it begins to cool and because the magma assimilates some of the crustal rock through which it ascends (country rock), and crustal rock tends to be high in silica. Silica content is thus the most important chemical criterion for classifying igneous rock.[7] The content of alkali metal oxides is next in importance.[11] About 65% of the Earth's crust by volume consists of igneous rocks. Of these, 66% are basalt and gabbro, 16% are granite, and 17% granodiorite and diorite. Only 0.6% are syenite and 0.3% are ultramafic. The oceanic crust is 99% basalt, which is an igneous rock of mafic composition. Granite and similar rocks, known as granitoids, dominate the continental crust.[12][13]

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- A sample of andesite (dark groundmass) with amygdaloidal vesicles filled with zeolite. Diameter of view is 8 cm.
- Andesite – Type of volcanic rock
- Alkali feldspar granite – Type of igneous rock rich in alkali feldspar
- Basalt – Magnesium- and iron-rich extrusive igneous rock
- 'A'ā – Molten rock expelled by a volcano during an eruption – Basaltic lava with a crumpled appearance
- Pāhoehoe – Molten rock expelled by a volcano during an eruption – Basaltic lava with a flowing, often ropy appearance
- Basaltic trachyandesite – Naming system for volcanic rocks
- Mugearite – Volcanic rock type

- Shoshonite – Potassium-rich variety of basaltic trachyandesite
- Basanite – A silica-undersaturated basalt
- Blairmorite – Rare porphyritic volcanic rock
- Boninite – Ultramafic extrusive rock high in both magnesium and silica
- Carbonatite – Igneous rock with more than 50% carbonate minerals
- Charnockite – Type of granite containing orthopyroxene
- Enderbite – Igneous rock of the charnockite series
- Dacite – Volcanic rock intermediate in composition between andesite and rhyolite
- Diabase, also known as dolerite – Type of igneous rock
- Diorite – Igneous rock type
- Napoleonite, also known as corsite – Variety of diorite with orbicular structure
- Dunite – Ultramafic and ultrabasic rock from Earth's mantle which is made of the mineral olivine
- Essexite – Igneous rock type
- Foidolite – Igneous rock rich in feldspathoid minerals
- Gabbro – Coarse-grained mafic intrusive rock
- Granite – Common type of intrusive, felsic, igneous rock with granular structure

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- Granodiorite – Type of coarse grained intrusive igneous rock
- Granophyre – Subvolcanic rock that contains quartz and alkali feldspar in characteristic angular intergrowths
- Harzburgite – Ultramafic mantle rock
- Hornblendite – Plutonic rock consisting mainly of the amphibole hornblende
- Hyaloclastite – Volcaniclastic accumulation or breccia
- Icelandite – Igneous rock type
- Ignimbrite – Type of volcanic rock
- Ijolite – Igneous rock consisting essentially of nepheline and augite
- Kimberlite – Igneous rock which sometimes contains diamonds
- Komatiite – Ultramafic mantle-derived volcanic rock
- Lamproite – Ultrapotassic mantle-derived volcanic or subvolcanic rock
- Lamprophyre – Ultrapotassic igneous rocks – An ultramafic,

ultrapotassic intrusive rock dominated by mafic phenocrysts
in a feldspar groundmass

- Latite – Type of volcanic rock – A silica-undersaturated form of andesite
- Lherzolite – an ultramafic and ultrabasic rock that is composed of olivine and pyroxene – An ultramafic rock, essentially a peridotite
- Monzogranite – A silica-undersaturated granite with <5% normative quartz
- Monzonite – Igneous intrusive rock with low quartz and equal plagioclase and alkali feldspar – a plutonic rock with <5% normative quartz
- Nepheline syenite – Holocrystalline plutonic rock – A silica-undersaturated plutonic rock of nepheline and alkali feldspar
- Nephelinite – Igneous rock made up almost entirely of nepheline and clinopyroxene – A silica-undersaturated plutonic rock with >90% nepheline
- Norite – igneous rock – A hypersthene-bearing gabbro
- Obsidian – Naturally occurring volcanic glass
- Pegmatite – Igneous rock with very large interlocked crystals
- Peridotite – Coarse-grained ultramafic igneous rock type
- Phonolite – Uncommon extrusive rock – A silica-undersaturated volcanic rock; essentially similar to nepheline syenite
- Phono tephrite – strongly alkaline volcanic rock with a composition between phonolite and tephrite – A volcanic rock with a composition between phonolite and tephrite
- Picrite – Variety of high-magnesium basalt that is very rich in the mineral olivine – An olivine-bearing basalt

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- Porphyry – Textural form of igneous rock with large grained crystals in a fine matrix
- Pumice – Light colored highly vesicular volcanic rock
- Pyroxenite – Igneous rock - a coarse grained plutonic rock composed of >90% pyroxene
- Quartz diorite – Igneous, plutonic rock – A diorite with >5% modal quartz
- Quartz monzonite – Type of igneous rock – An intermediate plutonic rock, essentially a monzonite with 5–10% modal quartz
- Quartzolite – Extremely rare igneous rock made mostly of

quartz - An intrusive rock composed mostly of quartz

□ Rhyodacite - Volcanic rock rich in silica and low in alkali

metal oxides - A felsic volcanic rock which is intermediate between a rhyolite and a dacite

□ Rhyolite - Igneous, volcanic rock, of felsic (silica-rich) composition

□ Comendite - Hard, peralkaline igneous rock, a type of light blue grey rhyolite

□ Pantellerite - Peralkaline rhyolite type of volcanic rock

□ Scoria - Dark vesicular volcanic rock

□ Shonkinite - Intrusive igneous rock - a plutonic rock

□ Sovite - igneous rock - A coarse-grained carbonatite rock

□ Syenite - Intrusive igneous rock - A plutonic rock dominated by orthoclase feldspar; a type of granitoid

□ Tachylite - Form of basaltic volcanic glass - Essentially a basaltic glass

□ Benmoreite - Silica-undersaturated volcanic rock of intermediate composition - sodic trachyandesite

□ Trachybasalt - A volcanic rock with a composition between basalt and trachyte

□ Hawaiite - Volcanic rock - a sodic type of trachybasalt, typically formed by ocean island (hot spot) volcanism

□ Trachyte - Extrusive igneous rock - A silica-undersaturated volcanic rock; essentially a feldspathoid-bearing rhyolite

□ Troctolite - Igneous rock - A plutonic ultramafic rock containing olivine, pyroxene and plagioclase

□ Tuff - Rock consolidated from volcanic ash

□ Websterite - ultramafic and ultrabasic rock - A type of pyroxenite, composed of clinopyroxene and orthopyroxene

□ Wehrlite - Ultramafic rock - An ultramafic plutonic or cumulate rock, a type of peridotite, composed of olivine and clinopyroxene

2.2 Describe rock textures.

Texture

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Gabbro specimen showing phaneritic texture, from Rock Creek Canyon, eastern Sierra Nevada, California

Rock microstructure

Texture is an important criterion for the naming of volcanic rocks. The texture of volcanic rocks, including the size, shape, orientation, and distribution of mineral grains and the inter

grain relationships, will determine whether the rock is termed a tuff, a pyroclastic lava or a simple lava. However, the texture is only a subordinate part of classifying volcanic rocks, as most often there needs to be chemical information gleaned from rocks with extremely fine-grained groundmass or from airfall tuffs, which may be formed from volcanic ash. Textural criteria are less critical in classifying intrusive rocks where the majority of minerals will be visible to the naked eye or at least using a hand lens, magnifying glass or microscope. Plutonic rocks also tend to be less texturally varied and less prone to showing distinctive structural fabrics. Textural terms can be used to differentiate different intrusive phases of large plutons, for instance porphyritic margins to large intrusive bodies, porphyry stocks and subvolcanic dikes. Mineralogical classification is most often used to classify plutonic rocks. Chemical classifications are preferred to classify volcanic rocks, with phenocryst species used as a prefix, e.g., "olivine-bearing picrite" or "orthoclase-phyric rhyolite."

2.3 Classify rocks on the basis of their composition and texture.

The IUGS recommends classifying igneous rocks by their mineral composition whenever possible. This is straightforward for coarse-grained intrusive igneous rock, but may require examination of thin sections under a microscope for fine-grained volcanic rock, and may be impossible for glassy volcanic rock. The rock must then be classified chemically. [18]

Mineralogical classification of an intrusive rock begins by determining if the rock is ultramafic, a carbonatite, or a lamprophyre. An ultramafic rock contains more than 90% of iron- and magnesium-rich minerals such as hornblende,

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pyroxene, or olivine, and such rocks have their own classification scheme. Likewise, rocks containing more than 50% carbonate minerals are classified as carbonatites, while lamprophyres are rare ultrapotassic rocks. Both are further classified based on detailed mineralogy. [19]

In the great majority of cases, the rock has a more typical mineral composition, with significant quartz, feldspars, or feldspathoids. Classification is based on the percentages of quartz, alkali feldspar, plagioclase, and feldspathoid out of the total fraction of the rock composed of these minerals, ignoring

all other minerals present. These percentages place the rock somewhere on the QAPF diagram, which often immediately determines the rock type. In a few cases, such as the diorite-gabbro-anorthite field, additional mineralogical criteria must be applied to determine the final classification. [19]

Where the mineralogy of an volcanic rock can be determined, it is classified using the same procedure, but with a modified QAPF diagram whose fields correspond to volcanic rock types. [19]

Chemical classification and petrology

Total alkali versus silica classification scheme (TAS) as proposed in Le Maitre's 2002 *Igneous Rocks – A classification and glossary of terms*[20] Blue area is roughly where alkaline rocks plot; yellow area is where subalkaline rocks plot.

When it is impractical to classify a volcanic rock by mineralogy, the rock must be classified chemically.

There are relatively few minerals that are important in the formation of common igneous rocks, because the magma from which the minerals crystallize is rich in only certain elements: silicon, oxygen, aluminium, sodium, potassium, calcium, iron, and magnesium. These are the elements that combine to form the silicate minerals, which account for over ninety percent of all igneous rocks. The chemistry of igneous rocks is expressed differently for major and minor elements and for trace

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elements. Contents of major and minor elements are conventionally expressed as weight percent oxides (e.g., 51% SiO₂, and 1.50% TiO₂). Abundances of trace elements are conventionally expressed as parts per million by weight (e.g., 420 ppm Ni, and 5.1 ppm Sm). The term "trace element" is typically used for elements present in most rocks at abundances less than 100 ppm or so, but some trace elements may be present in some rocks at abundances exceeding 1,000 ppm. The diversity of rock compositions has been defined by a huge mass of analytical data—over 230,000 rock analyses can be accessed on the web through a site sponsored by the U. S. National Science Foundation (see the External Link to EarthChem).[citation needed]

The single most important component is silica, SiO₂, whether occurring as quartz or combined with other oxides as feldspars or other minerals. Both intrusive and volcanic rocks are

grouped chemically by total silica content into broad categories.

□ Felsic rocks have the highest content of silica, and are predominantly composed of the felsic minerals' quartz and feldspar. These rocks (granite, rhyolite) are usually light colored, and have a relatively low density.

□ Intermediate rocks have a moderate content of silica, and are predominantly composed of feldspars. These rocks (diorite, andesite) are typically darker in colour than felsic rocks and somewhat more dense.

□ Mafic rocks have a relatively low silica content and are composed mostly of pyroxenes, olivines and calcic plagioclase. These rocks (basalt, gabbro) are usually dark coloured, and have a higher density than felsic rocks.

□ Ultramafic rock is very low in silica, with more than 90% of mafic minerals (komatiite, dunite).

□ This classification is summarized in the following table:

Composition

Ultramafi

Felsic Intermediate Mafic

Mode of c

(>63% (52% to 63% (45% to 52%

occurrence (<45%

SiO₂) SiO₂) SiO₂)

SiO₂)

Peridotit

Intrusive Granite Diorite Gabbro

e

Komatiit

Extrusive Rhyolite Andesite Basalt

e

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The percentage of alkali metal oxides (Na₂O plus K₂O) is second only to silica in its importance for chemically classifying volcanic rock. The silica and alkali metal oxide percentages are used to place volcanic rock on the TAS diagram, which is sufficient to immediately classify most volcanic rocks. Rocks in some fields, such as the trachyandesite field, are further classified by the ratio of potassium to sodium (so that potassic trachyandesites are latites and sodic trachyandesites are

Benmoreite). Some of the more mafic fields are further subdivided or defined by normative mineralogy, in which an idealized mineral composition is calculated for the rock based on its chemical composition. For example, basanite is distinguished from tephrite by having a high normative olivine content.

□ Other refinements to the basic TAS classification include:

Ultrapotassic – rocks containing molar $K_2O/Na_2O > 3$.

Peralkaline – rocks containing molar $(K_2O + Na_2O)/Al_2O_3 > 1$. [21]

Peraluminous – rocks containing molar $(K_2O + Na_2O + CaO)/Al_2O_3 < 1$. [21]

In older terminology, silica oversaturated rocks were called silicic or acidic where the SiO_2 was greater than 66% and the family term Quartzolite was applied to

the most silicic. A normative feldspathoid classifies a rock as silica-undersaturated; an example is nephelinite.

AFM ternary diagram showing the relative proportions of $Na_2O + K_2O$ (A for Alkali earth metals), $FeO + Fe_2O_3$ (F), and MgO (M) with arrows showing the path of chemical variation in tholeiitic and calc-alkaline series magmas

Magmas are further divided into three series:

The tholeiitic series – basaltic andesites and andesites.

The calc-alkaline series – andesites.

The alkaline series – subgroups of alkaline basalts and the rare, very high potassium-bearing (i.e. shoshonitic) lavas.

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The alkaline series is distinguishable from the other two on the TAS diagram, being higher in total alkali oxides for a given silica content, but the tholeiitic and calc-alkaline series occupy approximately the same part of the TAS diagram. They are distinguished by comparing total alkali with iron and magnesium content. [22]

These three-magma series occur in a range of plate tectonic settings. Tholeiitic magma series rocks are found, for example, at mid-ocean ridges, back-arc basins, oceanic islands formed by hotspots, island arcs and continental large igneous provinces. [23]

All three series are found in relatively close proximity to each other at subduction zones where their distribution is related to depth and the age of the subduction zone. The tholeiitic magma series is well represented above young subduction zones

formed by magma from relatively shallow depth. The calc-alkaline and alkaline series are seen in mature subduction zones, and are related to magma of greater depths. Andesite and basaltic andesite are the most abundant volcanic rock in island arc which is indicative of the calc-alkaline magmas. Some island arcs have distributed volcanic series as can be seen in the Japanese island arc system where the volcanic rocks change from tholeiite—calc-alkaline—alkaline with increasing distance from the trench.[24][25]

2.4 . Explain volcanic processes.

This article is about the process that forms volcanoes and igneous rocks. For the 18th century geological theory, see Plutonism.

Plate tectonics map with volcano locations indicated with red circles

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Volcanism, vulcanism or volcanicity is the phenomenon of eruption of molten rock (magma) onto the surface of the Earth or a solid-surface planet or moon, where lava, pyroclastic, and volcanic gases erupt through a break in the surface called a vent.[1] It includes all phenomena resulting from and causing magma within the crust or mantle of the body, to rise through the crust and form volcanic rocks on the surface. Magmas, that reach the surface and solidify, form extrusive landforms. [2] Volcanic activity is another result of the internal forces of the earth. It refers to the ejection of material from the interior of the earth. The physical feature through which the volcanic material rises towards the surface of the earth is called volcano. A volcano is an opening in the crust of the earth connected by a conduit to an underlying magma chamber through which molten lava, volcanic gases, steam and pyroclastic material is ejected.

The narrow opening of a volcano is called a vent. The upper part of the vent is often a cup or saucer shaped depression, called the crater. Sometimes the volcanic material is ejected through a long crack like opening. Such a volcano is called a fissure volcano. The material ejected through a volcano can be in the form of solid rocks or cinder, or molten rock material or magma. The solid material ejected through volcanic activity may be in the form of powder or ashes or in the form of small pebbles or large boulders. Besides the solid and liquid rock

material. Volcanoes spew out a large number of gases also. The rock material ejected during volcanic activity deposited on the surface of the earth is called lava.

It has been observed earlier that the temperature increases with increasing depth in the interior of earth. Therefore, the rocks in the interior should be in a

molten state. However, due to the high pressure of the overlying rocks, the materials in the interior of the earth behaves like solid substance. If the pressure of overlying rock strata on this material is released, it turns into molten material. Such a release of pressure may be caused due to faulting or fissures. The hot rock material turned into liquid (magma) rises to the surface through these cracks or fissures. Ejection of the material from the interior is called volcanic eruption. The

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material ejected to the surface gets cooled and solidifies thereby sealing the opening in the crust. Thus, the volcanic eruption lasts only a few days and volcanic activity is intermittent.

Volcanic processes

Non-viscous lava during an effusive eruption of Kilauea

Magma from the mantle or lower crust rises through the crust towards the surface. If magma reaches the surface, its behavior depends on the viscosity of the molten constituent rock.

Viscous (thick) magma produces volcanoes characterized by explosive eruptions, while non-viscous (runny) magma produce volcanoes characterized by effusive eruptions pouring large amounts of lava onto the surface.

In some cases, rising magma can cool and solidify without reaching the surface. Instead, the cooled and solidified igneous mass crystallizes within the crust to form an igneous intrusion.

As magma cools the chemicals in the crystals formed are effectively removed from the main mix of the magma (by a process known as fractional crystallization), so the chemical content of the remaining magma evolves as it solidifies slowly.

Fresh unevolved magma injections can remobilize more evolved magmas, allowing eruptions from more viscous magmas.

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Aspects of volcanism

Cross-section through a stratovolcano (vertical scale is exaggerated):

Large magma chamber

1. Bedrock
2. Conduit (pipe)
3. Base
4. Sill
5. Dike
6. Layers of ash emitted by the volcano
7. Flank
8. Layers of lava emitted by the volcano
9. Throat
10. Parasitic cone
11. Lava flow
12. Vent
13. Crater
14. Ash cloud

Volcanoes are places where magma reaches the planet's surface. The type of volcano depends on the location of the eruption and the consistency of the magma.

□ Earthquakes

Earthquakes are generally associated with plate tectonic activity, but some earthquakes are generated as a result of

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volcanic activity [3](though that itself is ultimately driven by the same forces).

□ Hydrothermal vents

These are formed where water interacts with volcanism. [4]

These include geysers, fumaroles, hot springs and mud pots, they are often used as a source of geothermal energy.[4]

□ Volcanic winter

The amount of gas and ash emitted by volcanic eruptions has a significant effect on the Earth's climate. Large eruptions correlate well with some significant climate change events.[5]

□ Forming rocks; Igneous rock and Metamorphic rock

When magma cools it solidifies and forms rocks. The type of rock formed depends on the chemical composition of the magma and how rapidly it cools. Magma that reaches the surface to become lava cools rapidly, resulting in rocks with small crystals such as basalt. Some of this magma may cool

extremely rapidly and will form volcanic glass (rocks without crystals) such as obsidian. Magma trapped below ground in thin intrusions cools more slowly than exposed magma and produces rocks with medium-sized crystals. Magma that remains trapped in large quantities below ground cools most slowly resulting in rocks with larger crystals, such as granite and gabbro.

Existing rocks that come into contact with magma may be melted and assimilated into the magma. Other rocks adjacent to the magma may be altered by contact metamorphism or metasomatism as they are affected by the heat and escaping or externally-circulating hydrothermal fluids.

□ Type of volcanicity

Active volcano

while there is no international consensus among volcanologists on how to define an "active" volcano, the USGS defines a volcano as "active" whenever subterranean indicators, such as earthquake swarms, ground inflation, or unusually high levels of carbon dioxide and/or sulfur dioxide are present. [56][57]

Dormant and reactivated

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Narcondam Island, India, is classified as a dormant volcano by the Geological Survey of India

The USGS defines a "dormant volcano" as Any volcano that is not showing any signs of unrest such as earthquake swarms, ground swelling, or excessive noxious gas emissions, but which shows signs that it could yet become active again. [57] Many dormant volcanoes have not erupted for thousands of years, but have still shown signs that they may be likely to erupt again in the future.[58][59]

□ Dormant volcano

In an article justifying the re-classification of Alaska's Mount Edgecumbe volcano from "dormant" to "active", volcanologists at the Alaska Volcano Observatory pointed out that the term "dormant" in reference to volcanoes has been deprecated over the past few decades and that "[t]he term "dormant volcano" is so little used and undefined in modern volcanology that the Encyclopedia of Volcanoes (2000) does not contain it in the glossaries or index,"[60] however the USGS still widely employs the term.

Previously a volcano was often considered to be extinct if there

were no written records of its activity. With modern volcanic activity monitoring techniques, it is now understood that volcanoes may remain dormant for a long period of time, and then become unexpectedly active again. For example, Yellowstone has a repose/recharge period of around 700,000 years, and Toba of around 380,000 years.[61] Vesuvius was described by Roman writers as having been covered with gardens and vineyards before its eruption of 79 CE, which destroyed the towns of Herculaneum and Pompeii.

It can sometimes be difficult to distinguish between an extinct volcano and a dormant (inactive) one. Pinatubo was an inconspicuous volcano, unknown to most people in the surrounding areas, and initially not seismically monitored before its unanticipated and catastrophic eruption of 1991. Two other examples of volcanoes which were once thought to be extinct, before springing back into eruptive activity were the long-dormant Soufrière Hills volcano on the island of Montserrat, thought to be extinct until activity resumed in 1995 (turning its capital Plymouth into a ghost town) and Fourpeaked 34

Mountain in Alaska, which, before its September 2006 eruption, had not erupted since before 8000 BCE.

□ Extinct volcano

Capulin Volcano National Monument in New Mexico, US

Extinct volcanoes are those that scientists consider unlikely to erupt again because the volcano no longer has a magma supply. Examples of extinct volcanoes are many volcanoes on the Hawaiian – Emperor seamount chain in the Pacific Ocean (although some volcanoes at the eastern end of the chain are active), Hohentwiel in Germany, Shiprock in New Mexico, US, Capulin in New Mexico, US, Zuid Wal volcano in the Netherlands, and many volcanoes in Italy such as Monte Vulture. Edinburgh Castle in Scotland is located atop an extinct volcano, which forms Castle Rock. Whether a volcano is truly extinct is often difficult to determine. Since "super volcano" calderas can have eruptive lifespans sometimes measured in millions of years, a caldera that has not produced an eruption in tens of thousands of years may be considered dormant instead of extinct.

□ Volcanic-alert level

The three common popular classifications of volcanoes can be

subjective and some volcanoes thought to have been extinct have erupted again. To help prevent people from falsely believing they are not at risk when living on or near a volcano, countries have adopted new classifications to describe the various levels and stages of volcanic activity. [62] Some alert systems use different numbers or colors to designate the different stages. Other systems use colors and words. Some systems use a combination of both.

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Decade volcanoes

Koryaksky volcano towering over Petropavlovsk-Kamchatsky on Kamchatka Peninsula, Far Eastern Russia

Main articles: Lists of volcanoes and Decade Volcanoes

The Decade Volcanoes are 16 volcanoes identified by the International Association of Volcanology and Chemistry of the Earth's Interior (IAVCEI) as being worthy of particular study in light of their history of large, destructive eruptions and proximity to populated areas. They are named Decade Volcanoes because the project was initiated as part of the United Nations-sponsored International Decade for Natural Disaster Reduction (the 1990s).

The 16 current Decade Volcanoes are:

1. Avachinsky-Koryaksky (grouped together), Kamchatka, Russia
2. Nevado de Colima, Jalisco and Colima, Mexico
3. Mount Etna, Sicily, Italy
4. Galeras, Nariño, Colombia
5. Mauna Loa, Hawaii, US
6. Mount Merapi, Central Java, Indonesia
7. Mount Nyiragongo, Democratic Republic of the Congo
8. Mount Rainier, Washington, US
9. Sakurajima, Kagoshima Prefecture, Japan
10. Santa Maria/Santiaguito, Guatemala
11. Santorini, Cyclades, Greece

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12. Taal Volcano, Luzon, Philippines
13. Teide, Canary Islands, Spain
14. Ulawun, New Britain, Papua New Guinea
15. Mount Unzen, Nagasaki Prefecture, Japan
16. Vesuvius, Naples, Italy

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2.5 DESCRIBE THE FORMATION AND CHARACTERISTICS OF PLUTONS.

□ FORMATION OF INTRUSIVE (PLUTONIC AND HYPABYSSAL)

Intrusive igneous rocks make up the majority of igneous rocks and are formed from magma that cools and solidifies within the crust of a planet. Bodies of intrusive rock are known as intrusions and are surrounded by pre-existing rock (called country rock). The country rock is an excellent thermal insulator, so the magma cools slowly, and intrusive rocks are coarse-grained (phaneritic). The mineral grains in such rocks can generally be identified with the naked eye. Intrusions can be classified according to the shape and size of the intrusive body and its relation to the bedding of the country rock into which it intrudes. Typical intrusive bodies are batholiths, stocks, laccoliths, sills and dikes. Common intrusive rocks are granite, gabbro, or diorite.

The central cores of major mountain ranges consist of intrusive igneous rocks. When exposed by erosion, these cores (called batholiths) may occupy huge areas of the Earth's surface.

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Intrusive igneous rocks that form at depth within the crust are termed plutonic (or abyssal) rocks and are usually coarse-grained. Intrusive igneous rocks that form near the surface are termed subvolcanic or hypabyssal rocks and they are usually much finer-grained, often resembling volcanic rock.[8] Hypabyssal rocks are less common than plutonic or volcanic rocks and often form dikes, sills,

laccoliths, lopoliths, or phacoliths.

A Plutonic Intrusive rock

Types of Intrusion:

1. Laccolith: concordant body with roughly flat base and convex top, usually with a feeder pipe below
2. small Dike
3. Batholith: a large irregular discordant intrusion
4. Dike: a relatively narrow tabular discordant body, often nearly vertical
5. Sill: a relatively thin tabular concordant body intruded along bedding planes
6. Volcanic pipe or volcanic neck: tubular, roughly vertical body that may have been a feeder vent for a volcano
7. Lopolith: concordant body with roughly flat top and a shallow convex base, may have a feeder dike or pipe below
8. Pluton emplacement

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9. Chonolith: an irregularly-shaped intrusion with a demonstrable base
10. Cupola: a dome-shaped projection from the top of a large subterranean intrusion
11. Phacolith: a concordant lens-shaped pluton that typically occupies the crest of an anticline or trough of a syncline
12. Stock: a smaller irregular discordant intrusive
13. Boss: a small stock

These are formed where magma pushes between existing rock, intrusions can be in the form of batholiths, dikes, sills and layered intrusions.

□ CHARACTERISTICS OF PLUTONIC INSTRUCTIVE IGNEOUS ROCK

1. Intrusive rocks are characterized by large crystal sizes, and as the individual crystals are visible, the rock is called phaneritic.[8]
2. There are few indications of flow in intrusive rocks, since their texture and structure mostly develops in the final stages of crystallization, when flow has ended. [9]
3. Contained gases cannot escape through the overlying strata, and these gases sometimes form cavities, often lined with large, well-shaped crystals.
4. These are particularly common in granites and their

presence is described as miarolitic texture.[10]

5. Because their crystals are of roughly equal size, intrusive rocks are said to be equigranular.[11]

6. Plutonic rocks are less likely than volcanic rocks to show a pronounced porphyritic texture, in which a first generation of large well-shaped crystals are embedded in a fine-grained ground-mass.

7. The minerals of each have formed in a definite order, and each has had a period of crystallization that may be very distinct or may have coincided with or overlapped the period of formation of some of the other ingredients.

8. Earlier crystals originated at a time when most of the rock was still liquid and are more or less perfect.

9. Later crystals are less regular in shape because they were compelled to occupy the spaces left between the already-formed crystals.

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10. The former case is said to be idiomorphic (or automorphic); the latter is xenomorphic.

11. Plutonic rocks form 7% of the Earth's current land surface. [13]

Intrusions vary widely, from mountain-range-sized batholiths to thin veinlike fracture fillings of aplite or pegmatite.

12. Hypabyssal rocks show structures intermediate between those of extrusive and plutonic rocks. They are very commonly porphyritic, vitreous, and sometimes even vesicular. In fact, many of them are petrologically indistinguishable from lavas of similar composition. [12][7] There are also many other characteristics that serve to distinguish plutonic from volcanic rock. For example, the alkali feldspar in plutonic rocks is typically orthoclase, while the higher-temperature polymorph, sanidine, is more common in volcanic rock. The same distinction holds for nepheline varieties. Leucite is common in lavas but very rare in plutonic rocks. Muscovite is confined to intrusions. These differences show the influence of the physical conditions under which crystallization takes place.[12]

Occurrences of Igneous intrusion

Dark dikes intruded into the country rock, Baranof Island, Alaska, United States

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0.3-UNDERSTAND PROCESSES THAT FORM METAMORPHIC ROCKS

3.1 Define metamorphic rock.

□ "METAMORPHIC ROCKS arise from the transformation of existing rock to new types of rock in a process called metamorphism. The original rock (protolith) is subjected to temperatures greater than 150 to 200 °C (300 to 400 °F) and, often, elevated pressure of 100 megapascals (1,000 bar) or more, causing profound physical or chemical changes. During this process, the rock remains mostly in the solid state, but gradually recrystallizes to a new texture or mineral composition.[1] The protolith may be an igneous, sedimentary, or existing metamorphic rock.

A type of metamorphic rock (disambiguation)
Quartzite,

CHARACTERISTIC OF METAMORPHIC ROCK

1. Metamorphic rocks make up a large part of the Earth's crust and form 12% of the Earth's land surface.[2]
 2. They are classified by their protolith, their chemical and mineral makeup, and their texture.
 3. They may be formed simply by being deeply buried beneath the Earth's surface, where they are subject to high temperatures and the great pressure of the rock layers above.
 4. They can also form from tectonic processes such as continental collisions, which cause horizontal pressure, friction, and distortion.
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5. Metamorphic rock can be formed locally when rock is heated by the intrusion of hot molten rock called magma from the Earth's interior.
 6. The study of metamorphic rocks (now exposed at the Earth's surface following erosion and uplift) provides information about the temperatures and pressures that occur at great depths within the Earth's crust.
 7. Some examples of metamorphic rocks are gneiss, slate, marble, schist, and quartzite. Slate[3] and quartzite[4] tiles are used in building construction. Marble is also prized for building construction[5] and as a medium for sculpture.[6] On

the other hand, schist bedrock can pose a challenge for civil engineering because of its pronounced planes of weakness.

[7]

3.2 Explain the formation of Metamorphic rock

□ FORMATION OF METAMORPHIC ROCK

Metamorphic rocks are formed by subjecting any rock type—sedimentary rock, igneous rock or another older metamorphic rock—to different temperature and pressure conditions than those in which the original rock was formed. This process is called metamorphism, meaning to "change in form". The result is a profound change in physical properties and chemistry of the stone. The original rock, known as the protolith, transforms into other mineral types or other forms of the same minerals, by recrystallization.[5] The temperatures and pressures required for this process are always higher than those found at the Earth's surface: temperatures greater than 150 to 200 °C and pressures greater than 1500 bars.[18] This occurs, for example, when continental plates collide.[19]: 31–33, 134–139 Metamorphic rocks compose 27.4% of the crust by volume. [13] The three major classes of metamorphic rock are based upon the formation mechanism. An intrusion of magma that heats the surrounding rock causes contact metamorphism—a temperature-dominated transformation. Pressure metamorphism occurs when sediments are buried deep under the ground; pressure is dominant, and temperature plays a smaller role. This is termed burial metamorphism, and it can result in rocks such as jade. Where both heat and pressure play a role, the mechanism is termed regional metamorphism. This is typically found in mountain-building regions.[7] Depending on the structure, metamorphic rocks

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are divided into two general categories. Those that possess a texture are referred to as foliated; the remainders are termed non-foliated. The name of the rock is then determined based on the types of minerals present. Schists are foliated rocks that are primarily composed of lamellar minerals such as micas. A gneiss has visible bands of differing lightness, with a common example being the granite gneiss. Other varieties of foliated rock include slates, phyllites, and mylonite. Familiar examples of non-foliated

metamorphic rocks include marble, soapstone, and serpentine. This branch contains quartzite—a metamorphosed form of sandstone—and hornfels.[7]

3.2 Classify metamorphic rocks on the basis of their texture (foliated on non-foliated)

TEXTURE IN METAMORPHIC ROCKS

The texture of a rock is defined by the small-scale structures seen in hand specimen and thin section, originating from the geometrical relationships of the minerals that constitute the rock. A strong foliation, produced by shearing, is the dominant feature of regionally metamorphosed rocks. Foliation can be seen from a distance as well as is often associated with isoclinal folds.

Because metamorphic transformation is a step-like process, a metamorphic rock may contain textures which:

- a) Were formed by igneous or sedimentary processes (relic textures), bedding or cross-bedding can survive low-medium grade metamorphism. Metamorphosed igneous rocks may show relics of porphyritic textures (the presence of phenocrysts). If a new mineral replaces an old mineral, e.g., old igneous minerals are replaced by new metamorphic minerals, then the new minerals is called a pseudomorph.
- b) Were formed by metamorphic processes. In regional and dynamic metamorphic rocks, these include new large metamorphic minerals set in a fine matrix, known as porphyroblasts, and strong foliations formed by preferred orientations of minerals or particles within the shear zone.
- c) Porphyroblasts and the minerals forming the matrix surrounding them may be elongate and have a preferred orientation.
- d) Deformation features such as microfolds may be present.

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Common features which make up the textures on contact metamorphic rocks include:

- ☐ Well formed (idoblastic) porphyroblasts that have a more random orientation.
- ☐ No obvious foliations
- ☐ The contact metamorphic minerals may overgrow pre-existing regional metamorphic minerals.

1.3 Describe the grade of Metamorphism as indicated

by foliated textures.

One of the most obvious things to note when looking at a metamorphic rock its texture (grain size, shape and orientation). Most rocks fit into one of two categories, foliated or non-foliated.

Common Metamorphic Rocks

Classification of Metamorphic Rock Parent Rock Common Minerals

Slate Fine grained < 1mm Shale, mudstone Quartz, clay minerals and feldspars

Schist medium grained 1- Shale, slate, basalt or Mica, chlorite, Foliated 5mm granite talc and quartz

Gneiss Coarse grained Shale, schist, granite, Quartz and >5mm sandstone and other feldspars rock types

Quartzite Sandstone Quartz

Marble Limestones, dolomite Calcite

Non-

Foliated

Anthracite coal Bituminous coal Crystalline carbon

0.4-Understand processes that form Sedimentary rocks

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4.1 Define sedimentary rocks and explain how they form.

□ SEDIMENTARY ROCKS are types of rock that are formed by the accumulation or deposition of mineral or organic particles at Earth's surface, followed by cementation. Sedimentation is the collective name for processes that cause these particles to settle in place. The particles that form a sedimentary rock are called sediment, and may be composed of geological detritus (minerals) or biological detritus (organic matter).

□ The geological detritus originated from weathering and erosion of existing rocks, or from the solidification of molten lava blobs erupted by volcanoes. The geological detritus is transported to the place of deposition by water, wind, ice or mass movement, which are called agents of denudation.

□ Biological detritus was formed by bodies and parts (mainly shells) of dead aquatic organisms, as well as their fecal mass, suspended in water and slowly piling up on the floor of

water bodies (marine snow). Sedimentation may also occur as dissolved minerals precipitate from water solution.

CHARACTERISTICS OF SEDIMENTARY

ROCK

1. The sedimentary rock cover of the continents of the Earth's crust is extensive (73% of the Earth's current land surface),

[1]

but sedimentary rock is estimated to be only 8% of the volume of the crust.[2]

2. Sedimentary rocks are only a thin veneer over a crust consisting mainly of igneous and metamorphic rocks.

3. Sedimentary rocks are deposited in layers as strata, forming a structure called bedding. Sedimentary rocks are often deposited in large structures

4. called sedimentary basins. Sedimentary rocks have also been found on Mars.

5. The study of sedimentary rocks and rock strata provides information about the subsurface that is useful for civil engineering, for example in the construction of roads, houses, tunnels, canals or other structures.

6. Sedimentary rocks are also important sources of natural resources including coal, fossil fuels, drinking water and ores.

7. The study of the sequence of sedimentary rock strata is the main source for an understanding of the Earth's history, including palaeogeography, paleoclimatology and the history of life.

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8. The scientific discipline that studies the properties and origin of sedimentary rocks is called sedimentology. Sedimentology is part of both geology and physical geography and overlaps partly with other disciplines in the Earth sciences, such as pedology, geomorphology, geochemistry and structural geology.

□ FORMATION OF SEDIMENTARY ROCK

Sedimentary rocks are formed at the earth's surface by the accumulation and cementation of fragments of earlier rocks, minerals, and organisms[14] or as chemical precipitates and organic growths in water (sedimentation). This process causes clastic sediments (pieces of rock) or organic particles (detritus) to settle and accumulate or for minerals to

chemically precipitate (evaporite) from a solution. The particulate matter then undergoes compaction and cementation at moderate temperatures and pressures (diagenesis).[5]: 265–280 [15]: 147–154 Before being deposited, sediments are formed by weathering of earlier rocks by erosion in a source area and then transported to the place of deposition by water, wind, ice, mass movement or glaciers (agents of denudation).[5] About 7.9% of the crust by volume is composed of sedimentary rocks, with 82% of those being shales, while the remainder consists of 6% limestone and 12% sandstone and arkoses.[13] Sedimentary rocks often contain fossils. Sedimentary rocks form under the influence of gravity and typically are deposited in horizontal or near horizontal layers or strata, and may be referred to as stratified rocks.[16] Sediment and the particles of clastic sedimentary rocks can be further classified by grain size. The smallest sediments are clay, followed by silt, sand, and gravel. Some systems include cobbles and boulders as measurements.[17]

4.2 Describe, name and classify common sedimentary rocks.

4.3 Classify clastic rocks on the basis of grain size and shape.

4.4 Classify chemical sedimentary rocks on the basis of composition.

4.5 Classify organic sedimentary rocks on the basis of origin.

0.5-Understand the porosity and permeability of the rocks

5.1 Define porosity and permeability.

5.2 Classify different porosity types.

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5.3 Describe the rates of porosity and permeability in different types of rocks.

0.6-Understand the types of aquifers as geological formations Based on their physical properties

6.1 Describe the various types of Classify rocks aquifers (aquifer, aquitard, aquifuge)

6.2 Describe subsurface zonation based on water saturation.

6.3 Describe the characteristics of aquifers.

6.4 Categorize rocks as aquifers or Aquicludes.

6.5 Describe the variety of subsurface aquifers being used globally.

0.7-Understand and interpret a table of formation

- 7.1 Describe criteria used to categorize rock units as Formations.
- 7.2 Explain the Era's of the Geological time Scale and their relative ages.
- 7.3 Describe effects of erosion, tilting and uplift on relationship between rock units.
- 0.8-Understand the difference between confirmed and non-confirmed aquifers
- 8.1 Recognize recharge and discharge areas of aquifers.
- 8.2 Describe the difference between confined and unconfined aquifers.
- 8.3 Determine water flow directions in confined and unconfined aquifer.
- 8.4 Use a potentiometric surface to predict water behavior in an artesian well.
- 0.9-Understand the effects of non-conformities on ground water movement
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- 9.1 Differentiate between 3 major unconformity types.
- 9.2 Describe processes that form unconformities.
- 9.3 Recognize different types of erosion surfaces and their effect on subsurface fluid flow.
- 1.0-Understand the processes that form geological structures and their effects on surface water and ground water
- 10.1 Define the Law of Original Horizontality
- 10.2 Differentiate types of stress and strain involved in the formation of rocks.
- 10.3 Identify processes that create faults and folds.
- 10.4 Identify faults and fold types.
- 10.5 Interpret strike, dip, fold and fault symbols on maps.
- 10.6 Differentiate surface water drainage patterns and their relationship to underlying geological features.
- 10.7 Explain the effect of structural geology on groundwater aquifers.
- 1.1-Understand how to subsurface relationship developed between rock units and their influence on ground water flow
- 11.1 Identify terrestrial, marine and transitional depositional environment. 11.2 Explain Depositional environments
- depositional Differentiate between depositional environmental

based on sedimentary features.

11.3 Describe the aquifer potential of different depositional environments based on geological characteristics.

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1.2-Understand basic climatological factors.

12.1 Describe the difference between climate and weather.

MEANING OF CLIMATE

□ Climate is defined as the average weather conditions of a place over a long period of time (30 years). The weather of any given place changes from day to day while climate is comparatively constant over a long time.

□ Climate is the long-term weather pattern in a region, typically averaged over 30 years.[1][2] More rigorously, it is the mean and variability of meteorological variables over a time spanning from months to millions of years. Some of the meteorological variables that are commonly measured are temperature, humidity, atmospheric pressure, wind, and precipitation. In a broader sense, climate is the state of the components of the climate system, including the atmosphere, hydrosphere, cryosphere, lithosphere and biosphere and the interactions between them.[1] The climate of a location is affected by its latitude, longitude, terrain, altitude, land use and nearby water bodies and their currents. [3]

□ Climates can be classified according to the average and typical variables, most commonly temperature and precipitation. The most widely used classification scheme was the Köppen climate classification. The Thornthwaite system,[4] in use since 1948, incorporates evapotranspiration along with temperature and precipitation information and is used in studying biological diversity and how climate change affects it. Finally, the Bergeron and Spatial Synoptic Classification systems focus on the origin of air masses that define the climate of a region.

□ The Intergovernmental Panel on Climate Change (IPCC) 2001 glossary definition is as follows: Climate in a narrow sense is usually defined as the "average weather", or more rigorously, as the statistical description in terms of the mean and variability of relevant quantities over a period ranging from months to thousands or millions of years. The classical period is 30 years, as defined by the World Meteorological

Organization (WMO). These quantities are most often surface variables such as temperature, precipitation, and wind. Climate

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in a wider sense is the state, including a statistical description, of the climate system.[10]

□ The World Meteorological Organization (WMO) describes "climate normal" as "reference points used by climatologists to compare current climatological trends to that of the past or what is considered typical. A climate normal is defined as the arithmetic average of a climate element (e.g. temperature) over a 30-year period. A 30-year period is used as it is long enough to filter out any interannual variation or anomalies such as El Niño–Southern Oscillation, but also short enough to be able to show longer climatic trends."[11]

MEANING OF WEATHER

□ Weather is defined as the conditions of the lower atmosphere of a place over a short period. The weather of a place is always very brief and changes regularly e.g the weather of today could be sunny , rainy or cloudy.

□ Weather is the state of the atmosphere, describing for example the degree to which it is hot or cold, wet or dry, calm or stormy, clear or cloudy.[1] On Earth, most weather phenomena occur in the lowest layer of the planet's atmosphere, the troposphere,[2][3] just below the stratosphere. Weather refers to day-to-day temperature, precipitation, and other atmospheric conditions, whereas climate is the term for the averaging of atmospheric conditions over longer periods of time. [4] When used without qualification, "weather" is generally understood to mean the weather of Earth.

□ Weather is driven by air pressure, temperature, and moisture differences between one place and another. These differences can occur due to the Sun's angle at any particular spot, which varies with latitude. The strong temperature contrast between polar and tropical air gives rise to the largest scale atmospheric circulations:

□ the Hadley cell, the Ferrel cell, the polar cell, and the jet stream. Weather systems in the middle latitudes, such as extratropical cyclones, are caused by instabilities of the jet streamflow. Because Earth's axis is tilted relative to its orbital plane (called the ecliptic), sunlight is incident at different

angles at different times of the year. On Earth's surface, temperatures usually range $\pm 40^\circ\text{C}$ (-40°F to 104°F) annually. Over thousands of years, changes in Earth's orbit can affect the amount and distribution of solar energy received by

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Earth, thus influencing long-term climate and global climate change.

Surface temperature differences in turn cause pressure differences. Higher altitudes are cooler than lower altitudes, as most atmospheric heating is due to contact with the Earth's surface while radiative losses to space are mostly constant.

□ Weather forecasting is the application of science and technology to predict the state of the atmosphere for a future time and a given location. Earth's weather system is a chaotic system; as a result, small changes to one part of the system can grow to have large effects on the system as a whole. Human attempts to control the weather have occurred throughout history, and there is evidence that human activities such as agriculture and industry have modified weather patterns

Studying how the weather works on other planets has been helpful in understanding how weather works on Earth. A famous landmark in the Solar System, Jupiter's Great Red Spot, is an anticyclonic storm known to have existed for at least 300 years. However, the weather is not limited to planetary bodies. A star's corona is constantly being lost to space, creating what is essentially a very thin atmosphere throughout the Solar System. The movement of mass ejected from the Sun is known as the solar wind.

12.2 Describe different types of climates.

The Köppen climate classification is one of the most widely used climate classification systems. It was first published by German-Russian climatologist Wladimir Köppen (1846–1940) in 1884,[2][3] with several later modifications by Köppen, notably in 1918 and 1936.[4][5] Later, German climatologist Rudolf Geiger (1894–1981) introduced some changes to the classification system, which is thus sometimes called the Köppen–Geiger climate classification.[6][7]

The Köppen climate classification divides climates into five main climate groups, with each group being divided based on patterns of seasonal precipitation and temperature. The five

main groups
are A (tropical), B (arid), C (temperate), D (continental),
and E (polar). Each group and subgroup is represented by a
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letter. All climates are assigned a main group (the first letter).
All climates except for those in the E group are assigned a
seasonal precipitation subgroup (the second letter). For
example, Af indicates a tropical rainforest climate. The system
assigns a temperature subgroup for all groups other than those
in the A group, indicated by the third letter for climates
in B, C, D, and the second letter for climates in E. For
example, Cfb indicates an oceanic climate with warm summers
as indicated by the ending b. Climates are classified based on
specific criteria unique to each climate type. [8]

As Köppen designed the system based on his experience as a
botanist, his main climate groups are based on the types of
vegetation occurring in a given climate classification region. In
addition to identifying climates, the system can be used to
analyze ecosystem conditions and identify the main types of
vegetation within climates. Due to its association with the plant
life of a given region, the system is useful in predicting future
changes of plant life within that region. [1]

The Köppen climate classification system was modified further
within the Trewartha climate classification system in 1966
(revised in 1980). The Trewartha system sought to create a
more refined middle latitude climate zone, which was one of
the criticisms of the Köppen system (the climate group C was
too general).[9]: 200–1

Overview[edit]

Köppen climate classification scheme symbols
description table[1][8][10]

1st 2nd 3rd

f (Rainforest)

m (Monsoon)

w (Savanna, dry

A (Tropical)

winter)

s (Savanna, dry

summer)

B (Dry) W (Arid Desert) h (Hot)

S (Semi-Arid or k (Cold)

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steppe)

a (Hot summer)

C w (Dry winter)

b (Warm

(Temperate) f (No dry season) summer)

s (Dry summer)

c (Cold summer)

a (Hot summer)

D b (Warm

w (Dry winter)

(Continental f (No dry season) summer)

) c (Cold summer)

s (Dry summer)

d (Very cold

winter)

E (Polar) T (Tundra)

F (Ice cap)

The Köppen climate classification scheme divides climates into five main climate

groups: A (tropical), B (arid), C (temperate), D (continental),

and E (polar).[11] The second letter indicates the seasonal

precipitation type, while the third letter indicates the level of

heat.[12] Summers are defined as the 6-month period that is

warmer either from April–September and/or October–March

while winter is the 6-month period that is cooler. [1][10]

Group A: tropical climates

□ This type of climate has every month of the year with an average temperature of 18 °C (64.4 °F) or higher, with significant precipitation.[1][10]

□ Af = Tropical rainforest climate; average precipitation of at least 60 mm (2.4 in) in every month.

□ Am = Tropical monsoon climate; driest month (which nearly always occurs at or soon after the "winter" solstice for that side of the equator) with precipitation less than 60 mm (2.4 in), but at least .[1][10]

□ Aw or As = Tropical wet and dry or savanna climate; with the driest month having precipitation less than 60 mm (2.4 in) and less than [1][10]

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Group B: arid climates

- This type of climate is defined by little precipitation that does not fit the polar (EF or ET) criteria of no month with an average temperature greater than 10 °C (50 °F).
- The threshold in millimeters is determined by multiplying the average annual temperature in Celsius by 20, then adding:
 - 280 if 70% or more of the total precipitation is in the spring and summer months (April–September in the Northern Hemisphere, or October–March in the Southern), or
 - 140 if 30%–70% of the total precipitation is received during the spring and summer, or
 - 0 if less than 30% of the total precipitation is received during the spring and summer.
- If the annual precipitation is less than 50% of this threshold, the classification is BW (arid: desert climate); if it is in the range of 50%–100% of the threshold, the classification is BS (semi-arid: steppe climate).[1][10]

□ A third letter can be included to indicate temperature. Originally, h signified low-latitude climate (average annual temperature above 18 °C (64.4 °F)) while k signified middle-latitude climate (average annual temperature less than 18 °C), but the more common practice presently, especially in the United States, is to use h to mean the coldest month that has an average temperature above 0 °C (32 °F) (or –3 °C (27 °F)), with k denoting that at least one month's averages below 0 °C (or –3 °C (27 °F)). In addition, n is used to denote a climate characterized by frequent fog and H for high altitudes.[13][14][15]

- BWh = Hot desert climate
- BWk = Cold desert climate
- BSh = Hot semi-arid climate
- BSk = Cold semi-arid climate

Group C: temperate climates

- This type of climate has the coldest month averaging between 0 °C (32 °F)[10] (or –3 °C (27 °F))[8] and 18 °C

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(64.4 °F) and at least one month averaging above 10 °C (50 °F).[10][8] For the distribution of precipitation in locations that both satisfy a dry summer (Cs) and a dry winter (Cw), a location is considered to have a wet summer (Cw) when

more precipitation falls within the summer months than the winter months while a location is considered to have a dry summer (Cs) when more precipitation falls within the winter months.[10] This additional criterion applies to locations that satisfies both Ds and Dw as well.[10]

☐ Cfa = Humid subtropical climate; coldest month averaging above 0 °C (32 °F) (or –3 °C (27 °F)), at least one month's average temperature above 22 °C (71.6 °F), and at least four months averaging above 10 °C (50 °F). No significant precipitation difference between seasons (neither the abovementioned set of conditions fulfilled). No dry months in the summer.

☐ Cfb = Temperate oceanic climate or subtropical highland climate; coldest month averaging above 0 °C (32 °F) (or –3 °C (27 °F)), all months with average temperatures below 22 °C (71.6 °F), and at least four months averaging above 10 °C (50 °F). No significant precipitation difference between seasons (neither the abovementioned set of conditions fulfilled).

☐ Cfc = Subpolar oceanic climate; coldest month averaging above 0 °C (32 °F) (or –3 °C (27 °F)) and 1–3 months averaging above 10 °C (50 °F). No significant precipitation difference between seasons (neither the abovementioned set of conditions fulfilled).

☐ Cwa = Monsoon-influenced humid subtropical climate; coldest month averaging above 0 °C (32 °F) (or –3 °C (27 °F)), at least one month's average temperature above 22 °C (71.6 °F), and at least four months averaging above 10 °C (50 °F). At least ten times as much rain in the wettest month of summer as in the driest month of winter (alternative definition is 70% or more of average annual precipitation is received in the warmest six months).

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☐ Cwb = Subtropical highland climate or Monsoon-influenced temperate oceanic climate; coldest month averaging above 0 °C (32 °F) (or –3 °C (27 °F)), all months with average temperatures below 22 °C (71.6 °F), and at least four months averaging above 10 °C (50 °F). At least ten times as much rain in the wettest month of summer as in the driest month of winter (an alternative definition is 70% or more of average annual precipitation received in the warmest six months).

☐ Cwc = Cold subtropical highland climate or Monsoon-influenced subpolar oceanic climate; coldest month averaging above 0 °C (32 °F) (or –3 °C (27 °F)) and 1–3 months averaging above 10 °C (50 °F). At least ten times as much rain in the wettest month of summer as in the driest month of winter (alternative definition is 70% or more of average annual precipitation is received in the warmest six months).

☐ Csa = Hot-summer Mediterranean climate; coldest month averaging above 0 °C (32 °F) (or –3 °C (27 °F)), at least one month's average temperature above 22 °C (71.6 °F), and at least four months averaging above 10 °C (50 °F). At least three times as much precipitation in the wettest month of winter as in the driest month of summer, and the driest month of summer receives less than 40 mm (1.6 in).[1]

☐ Csb = Warm-summer Mediterranean climate; coldest month averaging above 0 °C (32 °F) (or –3 °C (27 °F)), all months with average temperatures below 22 °C (71.6 °F), and at least four months averaging above 10 °C (50 °F). At least three times as much precipitation in the wettest month of winter as in the driest month of summer, and the driest month of summer receives less than 40 mm (1.6 in).[1]

☐ Csc = Cold-summer Mediterranean climate; coldest month averaging above 0 °C (32 °F) (or –3 °C (27 °F)) and 1–3 months averaging above 10 °C (50 °F). At least three times as much precipitation in the wettest month of winter as in the driest month of summer, and the driest month of summer receives less than 40 mm (1.6 in).[1]

Group D: continental climates

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☐ This type of climate has at least one month averaging below 0 °C (32 °F) (or –3 °C (27 °F)) and at least one month averaging above 10 °C (50 °F).[10][8]

☐ Dfa = Hot-summer humid continental climate; coldest month averaging below 0 °C (32 °F) (or –3 °C (27 °F)), at least one month's average temperature above 22 °C (71.6 °F), and at least four months averaging above 10 °C (50 °F). No significant precipitation difference between seasons (neither the abovementioned set of conditions fulfilled).

☐ Dfb = Warm-summer humid continental climate; coldest month averaging below 0 °C (32 °F) (or –3 °C (27 °F)), all

months with average temperatures below 22 °C (71.6 °F), and at least four months averaging above 10 °C (50 °F). No significant precipitation difference between seasons (neither the abovementioned set of conditions fulfilled).

□ Dfc = Subarctic climate; coldest month averaging below 0 °C (32 °F) (or –3 °C (27 °F)) and 1–3 months averaging above 10 °C (50 °F). No significant precipitation difference between seasons (neither the abovementioned set of conditions fulfilled).

□ Dfd = Extremely cold subarctic climate; coldest month averaging below –38 °C (–36.4 °F) and 1–3 months averaging above 10 °C (50 °F). No significant precipitation difference between seasons (neither the abovementioned set of conditions fulfilled).

□ Dwa = Monsoon-influenced hot-summer humid continental climate; coldest month averaging below 0 °C (32 °F) (or –3 °C (27 °F)), at least one month's average temperature above 22 °C (71.6 °F), and at least four months averaging above 10 °C (50 °F). At least ten times as much rain in the wettest month of summer as in the driest month of winter (alternative definition is 70% or more of average annual precipitation is received in the warmest six months).

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□ Dwb = Monsoon-influenced warm-summer humid continental climate; coldest month averaging below 0 °C (32 °F) (or –3 °C (27 °F)), all months with average temperatures below 22 °C (71.6 °F), and at least four months averaging above 10 °C (50 °F). At least ten times as much rain in the wettest month of summer as in the driest month of winter (alternative definition is 70% or more of average annual precipitation is received in the warmest six months).

□ Dwc = Monsoon-influenced subarctic climate; coldest month averaging below 0 °C (32 °F) (or –3 °C (27 °F)) and 1–3 months averaging above 10 °C (50 °F). At least ten times as much rain in the wettest month of summer as in the driest month of winter (alternative definition is 70% or more of average annual precipitation is received in the warmest six months).

□ Dwd = Monsoon-influenced extremely cold subarctic climate; coldest month averaging below –38 °C (–36.4 °F) and 1–3 months averaging above 10 °C (50 °F). At least ten times as

much rain in the wettest month of summer as in the driest month of winter (alternative definition is 70% or more of average annual precipitation is received in the warmest six months).

□ Dsa = Mediterranean-influenced hot-summer humid continental climate; coldest month averaging below 0 °C (32 °F) (or –3 °C (27 °F)), average temperature of the warmest month above 22 °C (71.6 °F) and at least four months averaging above 10 °C (50 °F). At least three times as much precipitation in the wettest month of winter as in the driest month of summer, and the driest month of summer receives less than 30 mm (1.2 in).

□ Dsb = Mediterranean-influenced warm-summer humid continental climate; coldest month averaging below 0 °C (32 °F) (or –3 °C (27 °F)), average temperature of the warmest month below 22 °C (71.6 °F) and at least four months averaging above 10 °C (50 °F). At least three times as much precipitation in the wettest month of winter as in the driest month of summer, and the driest month of summer receives less than 30 mm (1.2 in)

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□ Dsc = Mediterranean-influenced subarctic climate; coldest month averaging below 0 °C (32 °F) (or –3 °C (27 °F)) and 1–3 months averaging above 10 °C (50 °F). At least three times as much precipitation in the wettest month of winter as in the driest month of summer, and the driest month of summer receives less than 30 mm (1.2 in).

□ Dsd = Mediterranean-influenced extremely cold subarctic climate; coldest month averaging below –38 °C (–36.4 °F) and 1–3 months averaging above 10 °C (50 °F). At least three times as much precipitation in the wettest month of winter as in the driest month of summer, and the driest month of summer receives less than 30 mm (1.2 in).

Group E: polar and alpine climates

□ This type of climate has every month of the year with an average temperature below 10 °C (50 °F).[1][10]

□ ET = Tundra climate; average temperature of warmest month between 0 °C (32 °F) and 10 °C (50 °F).[1][10]

□ EF = Ice cap climate; eternal winter, with all 12 months of the year with average temperatures below 0 °C (32 °F).[1][10]

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12.3 Describe the various apparatus used for measuring weather elements.

S/ ELEMENTS INSTRUMENTS FUNCTIONS/USE UNITS

NO

1. Temperature Thermometer Degree of Hotness Centigrade or coldness of a place Celsius (°C) or Fahrenheit (°F).

2. Rainfall Rain gauge Average rainfall Millimeter (mm)

3. Pressure Barometer High- and low- Centimeter atmospheric pressure millibar & (mm & cm)

4. Humidity Hygrometer Absolute, relative Degree, and specific Percentage humidity °C, % and g/m³

5. Wind Windvane, Wind speed Time (Hours)

6. Wind Cup Wind directions Direction anemometer (N, S, NE, NW & SE) & SW

7. Sunshine Stevenson Intensity of the Meters screen & sundial sun's radiant (M) hour and sunshine duration

8. cloud Oktas Formations, Meters height and (M) appearance

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1. MEASUREMENT OF TEMPERATURE: the temperature of a place is measured by an instrument called THERMOMETER.

□ DESCRIPTION: thermometer is a narrow glass tube filled with mercury or alcohol. It works on the principle that mercury and alcohol expand when heated and contract

when cooled. Temperature is commonly recorded in centigrade (0C) or Fahrenheit (0F) scale.

□ The freezing (lowest) point for centigrade scale is 0 0C while Fahrenheit is 320F.

□ The boiling point (highest) point for centigrade scale is 1000C While that of the Fahrenheit is 2120F.

□ CONVERSION FROM ONE SCALE TO ANOTHER

I. To obtain centigrade = (0F - 32)

1.8

Example: convert 50 F To Centigrade

0

= (50 - 32)

1.8

= 18 = 100C

1.8

Ans: 50 F = 10 C

0 0

II. To obtain Fahrenheit = (1.8 * 0C) + 320F

Example: Convert 160C to Fahrenheit

= (1.8 * 16) + 320F

= (28.8 + 32) = 60.80F

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Ans: 160C = 60.8

Worked Example 1.

Use the statistics in the table below to answer the question that follow.

Mon J F M A M J J A S O N D

th

Tem. 40 30 21 25 10 40 48 30 25 30 20 15

(00)

□ CALCULATE: using the table above

I. The month with the highest temperature.

II. The month with the lowest temperature.

III. Annual range of temperature.

IV. Annual temperature.

V. Mean annual temperature.

VI. Draw a divergence line graph to represent the statistics.

2. MEASUREMENT OF RAINFALL: rainfall and other forms of precipitation is measured by an instrument called RAINGUAGE.

□ DESCRIPTION:

- I. A Rainguage consist of a funnel which 13cm - 20cm (5 inches - 8 inches).
- II. A funnel leads to the glass bottle.
- III. The hole in the funnel is very small.
- IV. The glass bottle is placed inside the copper cylinder.
- V. All are placed in the metal container.

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- VI. There is also a graduated measuring cylinder.

□ HOW RAIN GUAGE IS USED

- I. Every day, the funnel is removed.
- II. Any water in the glass bottle is poured into calibrated measuring cylinder.
- III. The reading is taken at eye level.
- IV. The readings are added up for each month.
- V. Total rainfall for the twelve months of the year is summed up and this gives the total annual rainfall.

□ PRECAUTION TO BE TAKEN IN THE USE OF RAIN GUAGE

- I. Rainguage must be sited in an open space.
- II. It must be at least 25 -30 cm (12 - 10 inches) above the ground.
- III. It must be fixed firmly to avoid spilling.
- IV. Reading must be taken daily
- V. The area where the instrument is kept must be fenced.

Use the statistics in the table below to answer the question that follow.

Mont J F M A M J J A S O N D

h

Rainf 50 40 20 25 10 15 80 30 5 90 20 60

all

cm3

□ CALCULATE: using the table above

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- I. Mean monthly rainfall.
- II. Annual rainfall.
- III. Annual range of rainfall
- IV. Mean annual rainfall.
- V. Highest rainfall.
- VI. Lowest rainfall.

3. MEASUREMENT OF PRESSURE: The instrument used in measuring atmospheric pressure is called BAROMETER.

□ DESCRIPTION:

I. Mercury barometer consists of a beaker in which a glass tube is dipped, and both contain mercury. Another instrument used for measuring pressure is the ANERIOD BAROMETER. But a modified aneroid barometer used in airplanes is called ALTIMETER.

II. Pressure is measured in unit of force called millibars.

III. A normal atmospheric pressure shows a reading of 760mm (76cm) or 1013 millibars.

IV. A reading of 790mm is regarded as high pressure while 700mm is regarded as low pressure.

4. MEASUREMENT OF RELATIVE HUMIDITY: the instrument used for measuring relative humidity is called HYGROMETER.

□ DESCRIPTION:

I. The hygrometer consists of wet and dry bulb thermometer placed side by side in the Stevenson screen.

II. The dry bulb is an ordinary thermometer that measures the shade temperature.

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III. The wet bulb, is kept wet by a wick that is dipped in a container of distilled water, when the air is not saturated, evaporation, which produces a cooling effect, takes place from the moist wick.

IV. The wet bulb always shows a lower reading than the dry bulb.

V. The difference in the two readings, if high, indicates low humidity and if the low, indicates high humidity.

VI. If there is no difference in their readings, it means that the air is saturated and therefore, the relative humidity would be 100%.

□ EXAMPLE:

□ Dry bulb temperature = 30°C

□ Wet bulb temperature = 28°C

□ Depression in Wet bulb = $30 - 28 = 20^\circ\text{C}$

□ Therefore, the relative humidity = 90%

5. MEASUREMENT OF WIND DIRECTION: the direction of wind is measured by an instrument called WIND VANE.

□ DESCRIPTION:

- I. Wind vane is made up of two parts.
- II. One part is the arrow or vane on the top. Which is free to move with the prevailing wind.
- III. The other part is stationary and consists of a frame and four compass points.
- IV. The vane or arrow points to the direction from which the wind is blowing so that wind is named from the direction it blows e. g south west wind blows from south west direction
- V. Wind vane is located in an exposed position so that tall buildings and trees do not deflect the wind direction.

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6. MEASUREMENT OF WIND SPEED: The speed of wind is measured by an instrument called ANEMOMETER.

□ DESCRIPTION:

- I. This consists of three or four semi-circular cups attached to the ends of horizontal spokes mounted on a high vertical spindle, as wind blows., the cups rotate.
- II. The higher the speed of the wind, the greater the speed of the cups and vice-versa.
- III. Reading is taken twice daily e. g 9.00 hours and 25.00 hours.
- IV. It is recorded in a note book.
- V. It states the date of recording.
- VI. The record is signed by the observer.

□ HOW WIND SPEED IS MEASURED

- I. It is measured with anemometer.
- II. It is placed in an open space.
- III. Anemometer consists of four metal cups fixed to a metal arm.
- IV. The shaft is connected to a meter.
- V. The cups rotate freely on a vertical shaft as the wind blows.
- VI. The number of rotation is indicated on a meter to give the speed of the wind in kilometer per hour.

12.4 Describe the factors/ causes responsible for climate change. (Temperature, precipitation, wind, pressure etc.)

ELEMENTS OF WEATHER AND CLIMATE

1. Temperature: is a physical quantity that expresses

quantitatively the perceptions of hotness and coldness.

Temperature is measured with a thermometer.

Thermometers are calibrated in various temperature

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scales that historically have relied on various reference points and thermometric substances for definition. The most common scales are the Celsius scale with the unit symbol °C (formerly called centigrade), the Fahrenheit scale (°F), and the Kelvin scale (K), the latter being used predominantly for scientific purposes. The kelvin is one of the seven base units in the International System of Units (SI). Temperature is important in all fields of natural science, including physics, chemistry, Earth science, astronomy, medicine, biology, ecology, material science, metallurgy, mechanical engineering and geography as well as most aspects of daily life.

2. Pressure: Pressure on Earth varies with the altitude of the surface, so air pressure on mountains is usually lower than air pressure at sea level. Pressure varies smoothly from the Earth's surface to the top of the mesosphere. Although the pressure changes with the weather, NASA has averaged the conditions for all parts of the earth year-round. As altitude increases, atmospheric pressure decreases. One can calculate the atmospheric pressure at a given altitude.

[9]

Temperature and humidity also affect the atmospheric pressure. Pressure is proportional to temperature and inversely proportional to humidity. And it is necessary to know both of these to compute an accurate figure.

Atmospheric pressure varies widely on Earth, and these changes are important in studying weather and climate.

3. Rainfall: Rain is water droplets that have condensed from atmospheric water vapor and then fall under gravity. Rain is a major component of the water cycle and is responsible for depositing most of the fresh water on the Earth. It provides water for hydroelectric power plants, crop irrigation, and suitable conditions for many types of ecosystems.

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The major cause of rain production is moisture moving along

three-dimensional zones of temperature and moisture contrasts known as weather fronts. If enough moisture and upward motion is present, precipitation falls from convective clouds (those with strong upward vertical motion) such as cumulonimbus (thunder clouds) which can organize into narrow rainbands. In mountainous areas, heavy precipitation is possible where upslope flow is maximized within windward sides of the terrain at elevation which forces moist air to condense and fall out as rainfall along the sides of mountains. On the leeward side of mountains, desert climates can exist due to the dry air caused by downslope flow which causes heating and drying of the air mass. The movement of the monsoon trough, or intertropical convergence zone, brings rainy seasons to savannah climes.

The urban heat island effect leads to increased rainfall, both in amounts and intensity, downwind of cities. Global warming is also causing changes in the precipitation pattern globally, including wetter conditions across eastern North America and drier conditions in the tropics. Antarctica is the driest continent. The globally averaged annual precipitation over land is 715 mm (28.1 in), but over the whole Earth, it is much higher at 990 mm (39 in). Rainfall is measured using rain gauges. Rainfall amounts can be estimated by weather radar.

4. Wind: wind is the natural movement of air or other gases relative to a planet's surface. Winds occur on a range of scales, from thunderstorm flows lasting tens of minutes, to local breezes generated by heating of land surfaces and lasting a few hours, to global winds resulting from the difference in absorption of solar energy between the climate zones on Earth. The two main causes of large-scale atmospheric circulation are the differential heating between the equator and the poles, and the rotation of the planet (Coriolis effect). In coastal areas the sea breeze/land breeze cycle can define local winds; in areas that have variable terrain, mountain and valley breezes can prevail. Winds are commonly classified by their spatial scale, their speed and direction, the forces that

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cause them, the regions in which they occur, and their effect.

Winds have various aspects: velocity (wind speed); the density of the gas involved; energy content, or wind energy. In meteorology, winds are often referred to according to their strength, and the direction from which the wind is blowing.

5. Humidity: Humidity is the concentration of water vapour present in the air. Water vapor, the gaseous state of water, is generally invisible to the human eye. [2] Humidity indicates the likelihood for precipitation, dew, or fog to be present. Humidity depends on the temperature and pressure of the system of interest. The same amount of water vapor results in higher relative humidity in cool air than warm air. A related parameter is the dew point. The amount of water vapor needed to achieve saturation increases as the temperature increases. As the temperature of a parcel of air decreases it will eventually reach the saturation point without adding or losing water mass. The amount of water vapor contained within a parcel of air can vary significantly. For example, a parcel of air near saturation may contain 28 g of water per cubic metre of air at 30 °C (86 °F), but only 8 g of water per cubic metre of air at 8 °C (46 °F).

Three primary measurements of humidity are widely employed: absolute, relative, and specific. Absolute humidity is expressed as either mass of water vapor per volume of moist air (in grams per cubic meter) [3] or as mass of water vapor per mass of dry air (usually in grams per kilogram).[4] Relative humidity, often expressed as a percentage, indicates a present state of absolute humidity relative to a maximum humidity given the same temperature. Specific humidity is the ratio of water vapor mass to total moist air parcel mass. Humidity plays an important role for surface life. A device used to measure humidity is called a hygrometer; one used to regulate it is called a humidistat, or sometimes hygostat.

6. Cloud: this is particles of droplet water in the upper atmosphere suspended in the air. Due to condensation of water vapour carried by rising air, clouds may give precipitation. Cloud reduces visibility and depletes solar insulation. Types of cloud include cirrus, cumulus and status. Tropospheric clouds exert numerous influences on Earth's troposphere and climate. First and foremost, they are the source of precipitation, thereby greatly influencing the

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distribution and amount of precipitation. Because of their differential buoyancy relative to surrounding cloud-free air, clouds can be associated with vertical motions of the air that may be convective, frontal, or cyclonic. The motion is upward if the clouds are less dense because condensation of water vapor releases heat, warming the air and thereby decreasing its density. This can lead to downward motion because lifting of the air results in cooling that increases its density. All of these effects are subtly dependent on the vertical temperature and moisture structure of the atmosphere and result in major redistribution of heat that affect the Earth's climate.

7. Sunshine: Sunlight is a portion of the electromagnetic radiation given off by the Sun, in particular infrared, visible, and ultraviolet light. On Earth, sunlight is scattered and filtered through Earth's atmosphere, and is obvious as daylight when the Sun is above the horizon. When direct solar radiation is not blocked by clouds, it is experienced as sunshine, a combination of bright light and radiant heat. When blocked by clouds or reflected off other objects, sunlight is diffused. Sources estimate a global average of between 164 watts to 340 watts[1] per square meter over a 24-hour day;[2] this figure is estimated by NASA to be about a quarter of Earth's average total solar irradiance.

The ultraviolet radiation in sunlight has both positive and negative health effects, as it is both a requisite for vitamin D3 synthesis and a mutagen.

Sunlight takes about 8.3 minutes to reach Earth from the surface of the Sun.[3] A photon starting at the center of the Sun and changing direction every time it encounters a charged particle would take between 10,000 and 170,000 years to get to the surface.[4]

Sunlight is a key factor in photosynthesis, the process used by plants and other autotrophic organisms to convert light energy, normally from the Sun, into chemical energy that can be used to synthesize carbohydrates and to fuel the organisms' activities.

Daylighting is the natural lighting of interior spaces by admitting sunlight. Solar irradiance is the solar energy

available from sunlight

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Researchers can measure the intensity of sunlight using a sunshine recorder, pyranometer, or pyrhelimeter. To calculate the amount of sunlight reaching the ground, both the eccentricity of Earth's elliptic orbit and the attenuation by Earth's atmosphere have to be taken into account.

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